

5

THEME: HUMAN HEALTH

TOPIC 5: OUR FOOD

Key words

- nourish
- proteins
- nutritional
- substance
- carbohydrates
- values

FOOD

Food is anything taken into the body to nourish it.

Food also means a substance that adds nutritional value in the body when taken in.

Nutrition is the process by which food is taken in and used by the body.

Learn, unlearn and relearn: It is wrong to say that food is anything/something/substance good to eat or drink and has values in the body because there is no element of nutritional value/ nourish in the definition. Additionally, even essential drugs add values in the body yet they are not food and some are drunk e.g., syrup. Thirdly, not every kind of food people eat is good! Some food may be half cooked, have dirt, some people like Chinese eat snake, Congolese eat caterpillars which many of the Ugandans cannot eat.

Ways in which food is taken into the body.

- Through eating.
- Through drinking.
- Through feeding tubes.

Reasons why people eat food.

- ✓ To have a healthy body.
- ✓ To make the body grow strong.
- ✓ To enable the body get energy.
- ✓ To promote proper body growth

Learn, unlearn and relearn: It is wrong to say that we eat food to be happy, it is a habit and for hospitality because they are not connected to the THEME "HUMAN HEALTH" instead they are just cultural perspectives.

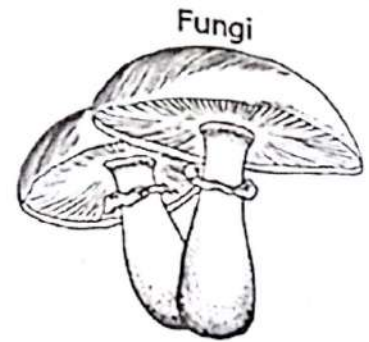
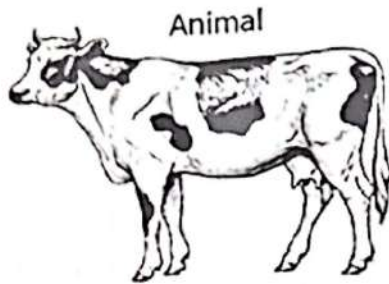
Ignore the 5Hs in teaching this topic since it is not reflected anywhere in the curriculum

Ways in which food is important in the body.

- ❖ Food provides energy to the body.
- ❖ Food builds the body.
- ❖ Food keeps the body healthy.
- ❖ Food promotes body growth.
- ❖ Food provides heat in the body.
- ❖ Food repairs damaged tissues.

Sources of food in the environment

Plants, animals and fungi



CLASSES OF FOOD.

Food is classified according to the chemical composition and nutrient value in the human body. Classes of food include; carbohydrates, proteins, vitamins, mineral salts and fats and oil

mnemonic used: **CPV FM**

Carbohydrates

These are foodstuffs that are broken down into the body to produce energy.

Sources of carbohydrates

- cassava ➤ sugar cane ➤ bread ➤ honey ➤ potatoes
- posho ➤ maize ➤ rice ➤ yams ➤ millet
- sorghum ➤ chapatti ➤ bans ➤ cakes



Bread



cassava



maize



Sweet potatoes

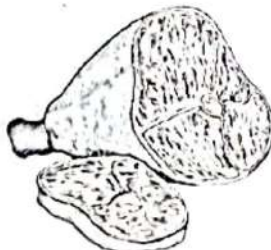
Proteins

These are foodstuffs that help to build the body.

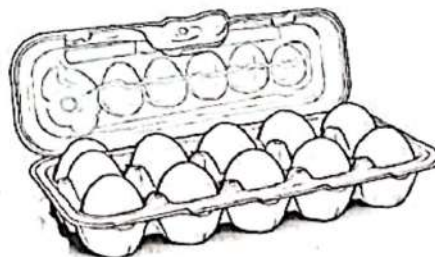
Proteins are grouped as animal proteins and plant proteins.

Sources of animal proteins

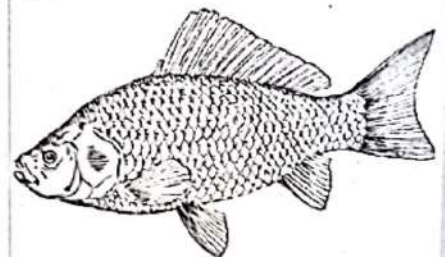
- eggs ▪ meat ▪ fish ▪ grasshoppers ▪ crickets



meat



eggs



fish

Sources of plant proteins

Beans, ground nuts, soya beans and cowpeas



Ground nuts



Cowpeas



Beans

Mushrooms are sources of fungi proteins.

Vitamins

These are foodstuffs that give health and protection to the body against diseases.

Sources of vitamins.

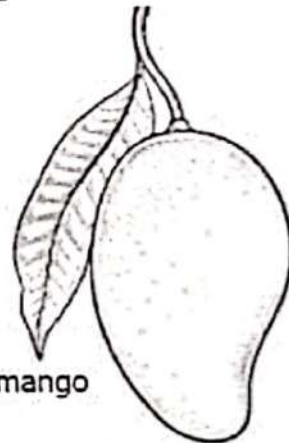
Oranges, vegetables, carrots, mangoes and tangerines



orange



cabbage



mango

Note well. Most fruits are eaten in raw form to maintain their vitamin value.

Most fruits are not cooked because cooking destroys their vitamin value.

Mineral salts

Mineral salts are foodstuffs that help to make bones strong, form tissues and blood in the body.

Sources of mineral salts.

Beans, meat, eggshells, tomatoes, liver, milk and green vegetables are rich in iron.

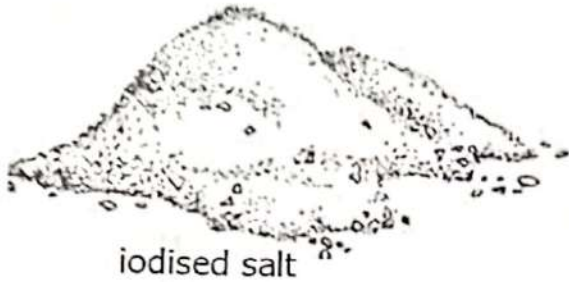
Iron is used in the formation of blood.

Smoked fish, meat, eggs, green vegetables, eggshells, cheese are rich in calcium.

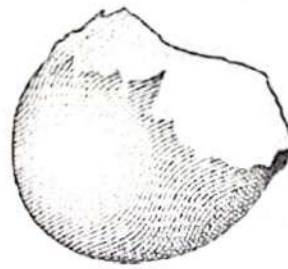
Calcium makes bones and teeth strong.

Iodised salt, seawater, cray fish, sea fish, shrimps and lobsters are rich in iodine.

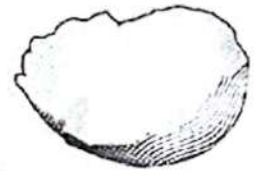
It helps in the proper working of the thyroid glands.



iodised salt



eggshell



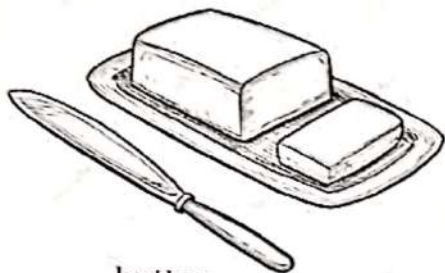
Fats and oils (lipids)

These are foodstuffs that provide heat in the body.

Fats are solids at room temperature while oils are liquids at room temperature. Lipids is the collective name for fats and oils.

Sources of fats and oils.

Lard, olive oil, simsim, palm oil, ghee, margarine, egg yolk, cheese and butter



butter



simsim



cheese

Learn, unlearn and relearn: Go food, Grow food and Glow food are not classes of food.

Food is classified according to chemical composition and nutrient value in the body. Not the resultant positive effect, so grow, go and glow are just descriptions of what the nutrients do in the human body.

Evaluation activity 5.1

1. (a) Give the meaning of the following terms:
 - (i) Food
 - (ii) Nutrition
- (b) State any **two** reasons why people eat food.
2. Give any **two** ways in which food is important in the human body.
3. Apart from proteins, name any **two** other classes of food commonly eaten by people.
4. What food value do you get when you eat posho at school regularly?
5. Name the class of food that helps in the hardening of human teeth and bones.
6. Give any **two** sources of animal proteins.

Key words.

- diet • dietary • nutrients • roughage • nutritional
- repair

FOOD VALUES

Food values are nutritional and dietary components of food.

Nutrients are substances which are essential for healthy growth.

Examples of food values and their uses in the human body.

Food values	Uses in the human body
proteins	▪ They build the body. ▪ They help in body growth. ▪ They repair worn out body tissues. ▪ They build body cells. ▪ They replace worn out body cells. ▪ They repair body cells.
carbohydrates	❖ They provide energy to the body.
Vitamins	• They help the body to remain healthy.
Mineral salts	▪ Calcium makes the bones and teeth strong. ▪ Iron helps in the formation of blood in the body.
Fats and oils	• They generate heat in the body. • They provide energy in the body.

Learn, unlearn and relearn: Proteins do not repair "worn out" body cells but repair body cells.

Importance of proteins to different vulnerable people.

Group of people.	Uses of proteins
Babies	• They promote proper body growth. • They build the body of the baby.
Elderly people	▪ They repair worn out body tissues. ▪ They build body cells. ▪ They replace worn out body cells
Pregnant mothers	▪ They repair worn out body tissues. ▪ They build body cells. ▪ They replace worn out body cells

 *Points to note: carbohydrates, fats and oils and proteins provide energy in the body.*

Did you know? Fats and oils provide more energy to the human body than carbohydrates.

Learn, unlearn and relearn: Roughages and water are not food values and classes of food. This is because they do not provide any nutritional values(nutrients) in the body.

BALANCED DIET

A balanced diet is a diet that contains all food nutrients in their right amounts. A diet refers to the overall pattern of food people consume over a period of time.

A meal is a single sitting of food eaten at a specific time.

Types/examples of meals

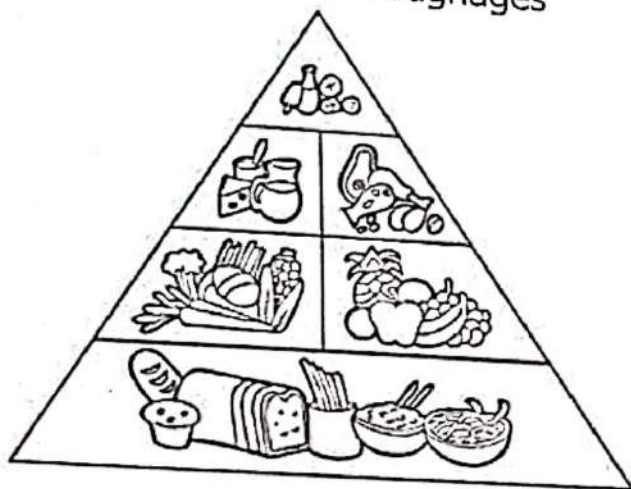
Breakfast, lunch, dinner and supper

Importance of a balanced diet

- ✓ It protects the body against malnutrition.
- ✓ It promotes proper body growth.

Components of a balanced diet

- carbohydrates
- fats and oils
- proteins
- roughages
- vitamins
- mineral salts
- water

**ROUGHAGES**

Roughages are plant-fibre materials in food.

They are also called dietary fibres. They help to prevent constipation by making the stool soft.

Importance of roughages in the human body.

They allow easy digestion of food. They add bulk to the diet.

Sources of roughages

- ☐ sweet potatoes
- ☐ bread
- ☐ wheat bran
- ☐ green leafy vegetables
- ☐ unpolished cereals
- ☐ fresh fruits

WATER**Importance of water in the human body.**

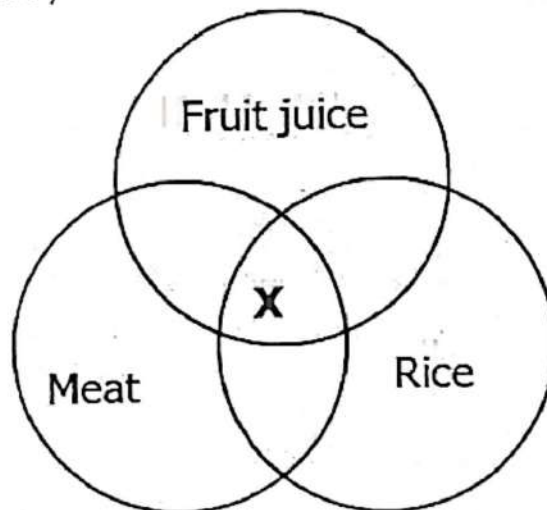
- ✓ It aids digestion of food.
- ✓ It regulates body temperature.
- ✓ It lubricates joints.
- ✓ It removes wastes like sweat and urine.
- ✓ It transports nutrients and oxygen.
- ✓ It increases the volume of blood.

Sources of water in the human body

- fruit juice
- milk
- fresh fruits
- soup
- coffee tea
- soda

Evaluation activity 5.2

1. (a) What is meant by the term balanced diet?
 (b) Write down any **two** examples of food values needed by the human body to remain healthy.
 (c) State any **one** reason why it is important for a person to feed on a balanced diet.
2. Give any **one** use of proteins in the human body.
3. Name the class of food that helps in the repair of body cells.
4. Apart from carbohydrates, name any **one** other class of food that gives the body energy.
5. Give **one** reason why babies should be fed on more food rich in proteins than the elderly people.
6. Give **one** reason why fruits like mangoes are eaten in raw form.
7. The diagram below shows foodstuffs eaten by family members in a certain home. Study and use it to answer the questions that follow.



- (a) What does the part marked **X** represent in the diagram above?
- (b) How does the father from the family who is a builder benefit from eating rice in the diagram above?
- (c) Which foodstuff from the diagram above would you recommend the mother to give her children to prevent kwashiorkor?
- (d) Name the food nutrient that the family members get from drinking fruit juice shown on the foodstuff above.

Key words

- deficiency • marasmus • stunted • nutrients • kwashiorkor
- starvation

DEFICIENCY DISEASES

Deficiency diseases are diseases caused by lack of food nutrients in the human body.

Kwashiorkor**Additional information**

Sometimes children do not get enough proteins because;

- Protein rich foods are expensive.
- Ignorance on how to feed children properly.
- Giving children more carbohydrates than proteins.
- Shortage of food at home.

- Kwashiorkor is usually got by children who are no longer breastfeeding because they easily get carbohydrates but not proteins.

Children should be given more food rich in proteins than other food to help children get enough proteins for proper growth.

Kwashiorkor is common in children due to rapid use of proteins for body growth.

Disease	Signs/symptoms	Prevention of the disease.
Kwashiorkor It is caused by deficiency of proteins in the human body.	▪ Swollen moon-like face. ▪ Stunted growth. ▪ Swollen belly. ▪ Swollen hands and feet. ▪ The hair takes on reddish colour ▪ The skin cracks and becomes scaly.	Feed a child on food rich in proteins.

Note: When you press the skin of a child with kwashiorkor, it takes long to come back to its position.

Reason. The whole body is swollen.

Disease	Signs/symptoms	Prevention of the disease.
Marasmus It is caused by Deficiency of all nutrients in the human body.	• A child has a bony face. • A child is always hungry. • Swollen potbelly. • The child grows very thin. • Old man's face. • Wasted muscles. • Loss of body weight.	Feed a child on a balanced diet.

Note: A child who is affected by marasmus mostly do not have enough food to eat or the food eaten does not have the required nutrients needed by the body.



A child suffering from marasmus

VITAMIN DEFICIENCY DISEASES

Night blindness

This is a health condition in which a person is unable to see properly when there is little light. They may not see properly in the evening when it is getting dark or in a room which is not well lit. Night blindness is caused by lack of vitamin A in the human body.

Signs of night blindness.

- Poor night vision.
- Unhealthy skin.
- Sore eyes
- Dry eyes.
- Reduced night vision.

Prevention of night blindness

Feed on food rich in vitamin A.

Such food includes carrots, milk, liver, fish, spinach, kale, mangoes, red pepper and pumpkins

Scurvy

- Scurvy is a skin deficiency disease.
- It is caused by deficiency of vitamin C in the human body.
- A child who does not eat enough vegetables and fruits may get scurvy.
- This is because the child lacks enough vitamin C in the body.

Signs and symptoms of scurvy

- ❖ Bleeding gums.
- ❖ Swelling of gums.
- ❖ Dry/scaly skin.
- ❖ Poor healing of wounds.
- ❖ Pain in the joints.
- ❖ Weak teeth.
- ❖ Wounds on the skin.

Prevention of scurvy

Feeding on food rich in vitamin C.

Such food includes fresh fruits like mangoes, oranges, tangerines and leafy vegetables.

Beriberi

It is caused by deficiency of vitamin B1 in the human body.
It affects the brain.

Signs and symptoms

- Loss of appetite
- Decreased muscle formation.
- Difficulty in speaking.
- Swelling of the arms and legs.
- Tiredness.
- Heart failure.
- Loss of feelings in the limbs.
- Mental confusion.

Prevention of beriberi.

Feeding on food rich in vitamin B1.

Such food include; yeast and beans.

Pellagra

It is caused by deficiency of vitamin B3 in the human body.

Signs and symptoms of pellagra.

- ↓ Skin sores at the mouth and eyes.
- ↓ Rough scaly skin
- ↓ General body weakness.

Prevention of pellagra.

Feeding on food rich in vitamin B3.

Such food includes; chicken, salmon, tuna, brown rice, whole grains and legumes.

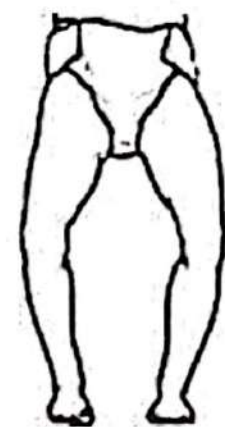
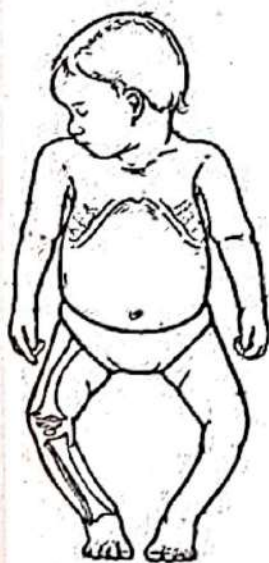
Learn, unlearn and relearn: Lack of vitamin B2 in a diet does not cause pellagra but it causes ariboflavinosis which is not part of primary concept.

Rickets

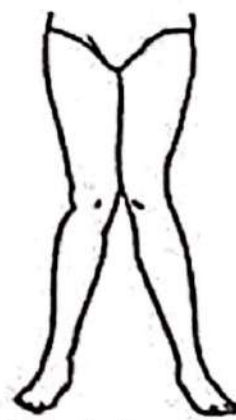
- Rickets is caused by deficiency of vitamin D in the human body.
- It is also caused by lack of calcium and phosphorus in a diet.
- Rickets affects bones in children.
- Vitamin D can be got from fatty food like fish, butter, milk and eggs.
- Sunlight enables the skin to make vitamin D.

Signs of rickets in children.

- Bow-shaped legs, Knock knee bones, Brittle bones and teeth, Poor teeth formation.



Bow-shaped legs



Knock-knee legs

Prevention of rickets in children.

Sunbathing in sunshine for a short time in the morning.

Feed the child on food rich in vitamin D.

Such food include; margarine, fatty fish, cod liver oil, egg yolks, beef liver, cheese and mushrooms.

Evaluation activity 5.3

- (a) What is meant by the term deficiency diseases?
(b) Mention any **two** examples of vitamin deficiency diseases.
(c) State the best way of preventing deficiency diseases.
- Which deficiency disease is caused by lack of vitamin B1 in a diet?
- (a) Name the disease which results from the deficiency of vitamin C in the human body.
(b) State any **two** signs of vitamin C deficiency in the human body.
(c) Give any **one** example of food which gives us vitamin C.
- In the table below, part **A** shows of deficiency diseases and their signs in part **B** of the table.

Part A: deficiency diseases	Part B: Signs
Scurvy	Reduced night vision.
Rickets	Heart failure.
Beriberi	Bow-shaped legs.
Night blindness	Bleeding gums.

Use the signs in part B of the table to match the deficiency diseases below.

- Scurvy
- Rickets
- Beriberi
- Night blindness

Key words

☐ goitre
☐ vital

☐ anaemia
☐ pale

☐ malnutrition

☐ nutrients

MINERAL DEFICIENCY DISEASES

Goitre

Goitre is a health condition in which one's glands in the throat swell. It is caused by deficiency of iodine in the human body.

Signs and symptoms of goitre	Prevention of goitre
Swollen neck.	Feeding on food rich in iodine. Such food includes; iodised salt, shrimps, sea fish ground water, egg yolk and cheese

Anaemia

Anaemia is a health condition in which a person lacks blood in the body. Anaemia is caused by deficiency of iron in the human body.

Other causes of anaemia

- Malaria infection. Malaria parasites destroy the red blood cells.

- Hookworm infestation. Hookworms feed on blood.
- Jigger infestation. Jiggers feed on blood.
- Excessive bleeding. This causes to loss of too much blood in the body.
- Bilharzia. Blood flukes feed on blood.

Signs and symptoms of anaemia	Prevention of goitre
• Pale skin • White fingernails. • Feeling tired and weak. • Pale gums. • Pale eyelids and the tongue.	❖ Feeding on food rich in iron. Such food include; Meat, eggs, liver, fish, spinach, ground nuts beans, peas and tomatoes. ❖ Swallowing iron supplement tablets Such tablets include; feosol, fergon, niferex.



A person suffering from goitre



A nurse caring for an anaemic person

MALNUTRITION

Malnutrition is a condition of poor health caused when the body does not get the right or required amount of nutrients.

Malnutrition comes from two words; **Mal** which means abnormal and **nutrition** meaning feeding.

Causes of malnutrition in children.

Cause/factor	How it causes malnutrition
Tapeworm infestation	Tapeworms feed on digested food.
Improper breastfeeding	Inadequate breastfeeding deprives infants of vital nutrients.
Food insecurity	It causes shortage of access to nutritious food.
Floods and drought	Disrupt food production and supply. Destroy food in the garden.
Poverty	It restricts access to nutritious food.

Cause/factor	How it causes malnutrition
Food taboos	Limit some vital nutrient intake.
Diseases like diarrhoea	Impair nutrient absorption and appetite for food.
Wars	They limit access to nutritious food. They disrupt food production.
Improper weaning	Introducing solid food too early or too late to the baby can result into nutrient deficiency.

Signs of malnutrition in children.

- Retarded growth.
- Body weakness
- Chronic tiredness
- Low body immunity
- Loss of body weight.

Effects of malnutrition to a child.

- ✓ It causes stunted growth in children.
- ✓ It lowers the immunity of the child.
- ✓ It prevents proper brain development.
- ✓ It causes chronic fatigue.
- ✓ It leads to loss of interest in work.



A malnourished child

OBESEITY

Obesity is a chronic condition caused by having excess fats in the body.

Factors that can lead to obesity

- Over eating.
- Lack of physical exercises.
- Eating a lot of fatty food.
- Medication like antidepressants.

Effects of obesity to human health.

- It can lead to high blood pressure.
- It can lead to diabetes.
- It leads to reduced life expectancy.
- It causes breathing complications.
- It may lead to death.
- It causes oedema.

Signs of good health in a child.

- ❖ Mental alertness.
- ❖ Good eyesight.
- ❖ Strong bones and teeth.
- ❖ Having energy.



An obese person

Evaluation activity 5.4

1. Give any **one** example of mineral deficiency disease in humans.
 2. Which food nutrient helps to prevent anaemia in children?
 3. What is meant by the term malnutrition?
 4. Give any **two** factors that can lead to malnutrition in children.
 5. How does tapeworm infestation lead to malnutrition in humans?
- The diagram below shows health condition in humans. Use it to answer questions **6** and **7**



6. Name the health condition in humans above.
7. State any **one** cause of the health condition above in humans.
8. State any **two** effects of malnutrition in children.
9. (a) Name the disease which results from the deficiency of iodine in the human body.
(b) State any **two** signs of iodine deficiency disease in humans.
(c) Give any **one** example of food which gives us iodine.

Key words:

- contaminate ▪ substance ▪ dysentery ▪ foodborne ▪ poisoning
- expire

FOOD CONTAMINATION

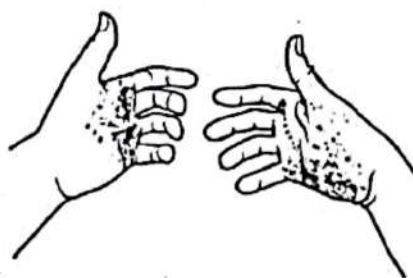
Food contamination is the presence of harmful substances in food that can make it unsafe to eat.

Substances that contaminate food.

- germs ➤ dust ➤ poison ➤ glass particles ➤ metal shards
- vectors

Ways in which food can get contaminated.

- ❖ Serving food with dirty hands.
- ❖ Leaving food uncovered.
- ❖ Serving food in dirty containers.
- ❖ Serving food from dirty places.
- ❖ Cooking food in dirty places and saucepans.
- ❖ Covering food with dirty containers.



Dirty hands



Uncovered food



Dirty plates

Dangers of eating contaminated food to human health

- It causes food poisoning.
- It can lead to death of a person.
- It makes one to suffer from foodborne diseases.
- It can lead to vomiting.
- It causes stomach cramps.

Food poisoning

Food poisoning is an illness caused to human health by eating contaminated food.

Causes of food poisoning

- Presence of germs in food.
- Putting toxins in food.
- Cross-contamination.
- Eating spoilt or expired food.
- Eating raw shellfish.
- Presence of vectors on food.
- Poor food hygiene.
- Undercooking food.
- Improper storage of food.
- Drinking unboiled milk or water

Ways of preventing food poisoning in human health.

- ✓ Eating steamed food.
- ✓ Cook food thoroughly.
- ✓ Store food at safe temperatures.
- ✓ Practice good food hygiene.
- ✓ Eat well washed fruits and vegetables.
- ✓ Refrigerate leftover food promptly.
- ✓ Check for expiry date on packed food.
- ✓ Clean kitchen surfaces daily.
- ✓ Ensure proper food packaging.
- ✓ Keep raw and cooked food separate.

Foodborne diseases

These are infections caused when a person eats contaminated food.

Examples of foodborne diseases

Diarrhoea, dysentery, typhoid, cholera, typhoid and tuberculosis

Ways in which food contamination can be prevented.

- ⚡ Covering food properly.
- ⚡ Serving food from clean places.
- ⚡ Preparing food in clean containers.
- ⚡ Proper sanitation
- ⚡ Proper disposal of rubbish.
- ⚡ Proper sensitisation of people about hygiene.
- ⚡ Serving food in clean plates.
- ⚡ Washing hands before touching food.
- ⚡ Covering food with clean lids.
- ⚡ Raising stalls above the ground.

Note well:

We wash hands before eating food to remove germs and dirt from hands.

We wash hands after eating food to remove food remains from hands.

Tuberculosis is a foodborne disease because it can spread through drinking unboiled milk from an infected cow.

FOOD HYGIENE

Food hygiene is the general cleanliness of food and things used when preparing food.

Ways of promoting food hygiene.

- Serving food in clean containers
- Serving food from clean places.
- Washing hands before touching food.
- Keeping food in clean places and plates.



Washing utensils



Clean kitchen



Washing food

Roles played by the school health club to ensure food hygiene at school.

- (i) Educating pupils on proper food handling.
- (ii) Displaying posters on food hygiene practices.
- (iii) Providing sanitation facilities near eating areas.
- (iv) Encouraging proper waste disposal near food areas.
- (v) Inspecting the school kitchen and dining areas regularly.
- (vi) Monitoring the hygiene of food vendors at school.

Importance of food hygiene to human health.

-It prevents food poisoning.

-It prevents foodborne diseases.

Evaluation activity 5.5

1. (a) What is meant by the term food contamination?
- (b) Mention any **two** ways in which food may get contaminated after it has been prepared.
- (c) State any **one** way in which you can prevent food from getting contaminated after serving it.
2. What name is given to an illness caused by eating contaminated food?
3. Write down any **three** substances that contaminate food.
4. State any **two** causes of food poisoning at home.
5. State any **two** ways in which contaminated food affects the health of humans.
6. Why should we wash our hands before eating food?
7. Give any **two** examples of foodborne diseases.
8. State any **one** way of controlling food poisoning at home.
9. Give any **two** roles played by school Health Club to ensure food hygiene at school.

Key words:

- refrigerate
- mould
- irradiate
- pasteurise
- preservative
- pickle

FOOD PRESERVATION

Food preservation is a way of keeping food safe for a long time without going bad.

Organisms like bacteria and fungi make food to go bad.

Changes in food that show that it has gone bad.

- Develops bad smell
- Unpleasant taste
- Change of colour
- Grows moulds
- Becomes sticky
- Bubbles form in food

Traditional/local methods of preserving food.

Sun-drying

This involves putting/spreading food in sunshine to dry.

Heat from the sun dehydrates food. /it drains moisture from food.

Sun-drying prevents food from rotting by dehydrating it.

Foodstuffs that are sun-dried include; fish, cassava, maize, millet, beans, sorghum.

Note. Wind also helps to preserve food by increasing the rate of evaporation in food.

Smoking

This involves using slow burning heat to dry food.

Heat from smoke dehydrates food.

Foodstuffs preserved by smoking are fish and meat.

Salting

This involves soaking food into salt solution or sprinkling salt on food.

Salt dehydrates food.

Examples of foodstuffs preserved by salting are meat and fish.

Note: These traditional methods of food preservation are recalled using 3Ss.



Advantages of traditional methods of food preservation.

- They are cheap and easy to use.
- They preserve food for long time.
- They are effective against bacteria.
- No additives are required.
- They add flavour in food e.g. salting.

Disadvantages of traditional methods of food preservation.

- They are time-consuming.
- They alter the taste of food.
- They easily contaminate food.
- Smoking pollutes air.
- They are weather dependent e.g. sun-drying.

Modern methods of preserving food.

Refrigeration

This involves keeping food in a fridge under cold conditions.

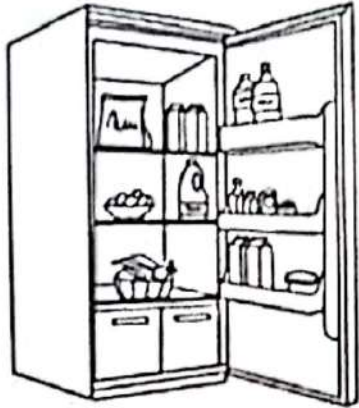
The low temperature in a fridge/refrigerator prevents the breeding of bacteria or moulds on food.

Foodstuffs preserved by refrigeration include; meat, fish, fruits, vegetables, milk, onions

Pickling

This involves soaking food in acids like vinegar (acetic acid)
It reduces the growth of bacteria in food.

Examples of foodstuffs preserved by pickling are cucumber, carrots, onions, eggs, peppers, lemons, watermelons, fish, meat, garlic, ginger and olives.



Refrigeration



Pickling

Irradiation

This is a process of exposing food to strong heat.

Radiant heat kills bacteria in food.

Examples of foodstuffs preserved by irradiation are; chicken, shellfish, vegetables

Sterilisation

Food like milk is heated at high temperature, cooled and sealed in sterile containers.

Bacteria are killed by maximum boiling.

Pasteurisation

Food like milk is heated at controlled temperature and allowed to cool.

Pasteurisation kills bacteria in milk.

Freezing

This involves keeping food at a very low temperature in order to preserve it.

The cold temperature prevents the breeding of bacteria in food.

How freezing is applied when preserving food.

- Boiling, cooling and then freezing vegetables.
- Freezing berries quickly individually before packaging.
- Using liquid nitrogen to freeze food like meat.
- Using household freezers to store meat, prepared meat and leftovers.

Using food preservatives.

They prevent bacteria and mould from breeding in food.

Examples of food preservatives

salt, sugar, local salt and vinegar, sodium benzoate.

Ice-glazing

This involves covering food with a thin layer of ice, which prevents dehydration and freezer burn.

Examples of foodstuffs ice-glazed are fish, shrimps and fruits.

Ice cubes are put in fresh fish.

Advantages of modern methods of preserving food.

- ❖ Longer shelf life.
- ❖ They are convenient to use.
- ❖ They are faster in killing bacteria.
- ❖ They maintain the nutritional value of food.
- ❖ Reduce food waste.
- ❖ They improve food safety.

Disadvantages of modern methods of preserving food.

- ❖ Some items used are not easily accessible e.g. vinegar.
- ❖ Some modern methods require electricity to work.
- ❖ They are expensive to use e.g. refrigeration
- ❖ The flavour in food can easily get lost.

Reasons why food should be preserved.

- ✓ To ensure food security.
- ✓ To add nutritional value to food.
- ✓ To keep food for future use.
- ✓ To increase on the lifespan of food.
- ✓ For easy transportation.
- ✓ To maintain the taste of food.

FOOD SECURITY

Food security is a situation in which there is enough food available all time for consumption.

Ways of promoting food security

- Proper storage of food.
- Preserving food.
- Growing drought resistant crops.
- Storing excess food.
- Growing more food crops.
- Early planting

Importance of food security.

-It prevents famine.

-It prevents malnutrition.

Problems faced by people when getting food.

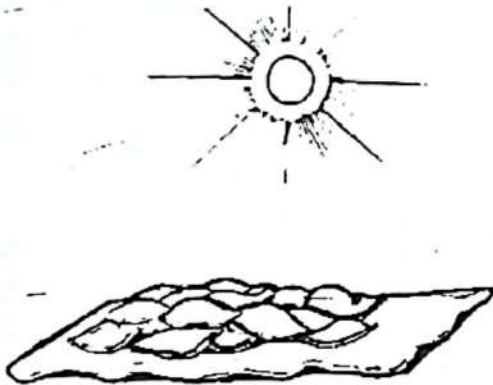
- (i) Contamination of food.
- (iii) Accidents along the way.
- (v) Low quality food.
- (vii) Failure to get fresh food.
- (ix) Overstayed food.
- (ii) High prices of food.
- (iv) Wrong labelling of food.
- (vi) Poor transport.
- (viii) Long distances to the market.

Evaluation activity 5.6

1. (a) Give the meaning of the following terms:
 - (i) Food preservation
 - (ii) Food security
- (b) State any **two** ways in which parents can ensure the availability of enough food at home.

Evaluation activity 5.6

2. The diagram below shows methods of preserving food. Study and use them to answer the questions that follow.

**W****X**

- Name the methods of preserving food marked **W** and **X**.
 - Give any **one** vegetable that can be preserved using the method marked **W**.
 - State any **one** advantage of using method marked **X** over method marked **W** when preserving food at home.
- How is sun-drying helpful in the preservation of food crops?
 - How does smoking control the effect of bacteria in food preservation?
 - Name any **one** organism that makes fresh meat go bad.
 - State any **three** ways of preventing fresh meat from going bad
 - State any **two** changes in food which show that food has gone bad.
 - Mention any **two** methods of preserving food that;
 - Do not require heat:
 - Keep food in cold conditions:

Key words:

▪ matoke ▪ mingle ▪ roast ▪ dish ▪ stew ▪ consumption

PREPARATION OF SIMPLE DISHES USING LOCAL FOODS

Food preparation is the way of making food ready for consumption

A dish is a combination of staple foods and sauce.

Examples of dishes that can be prepared at home

- mashed matoke • fried rice (pilau) • millet bread
- posho • boiled potatoes • roasted cassava
- chips • stew • salads

Methods that can be used to prepare simple dishes at home.

Method	Examples of food prepared
Boiling	Porridge, rice, potatoes, cassava, meat, milk tea
mingling	Millet bread, posho, cassava bread, matoke
frying	Meat, fish, rice, matoke

Method	Examples of food prepared
roasting	Meat, ground nuts, cassava, maize, fish
baking	Cakes, bread, bans, chapattis
steaming	Matoke, rice, meat, fish, vegetables
stewing	Matoke, meat



Boiling



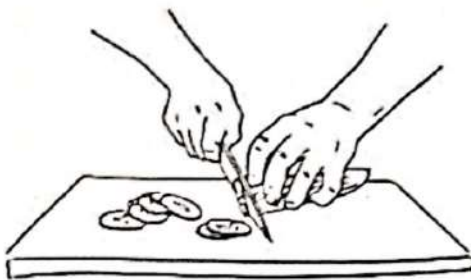
Mingling



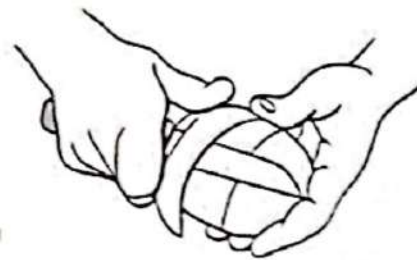
Roasting

Steps taken to prepare a meal at home.

Sorting food, washing food, peeling food, slicing food and chopping food



Slicing food



Peeling food



Washing food

How to Prepare some Dishes at Home

Preparing matoke mixed with groundnuts.

Materials needed:

- ☒ 16 fingers of matoke
- ☒ salt
- ☒ knife
- ☒ ¼ kg of pounded groundnuts
- ☒ water
- ☒ saucepan
- ☒ onions, tomatoes (one if big)

Steps taken:

- (i) Peel the 16 fingers of matoke and cut them into pieces.
- (ii) Place the matoke in a saucepan, add water, and a pinch of salt, then place it on the charcoal stove to cook.
- (iii) While matoke is cooking, grind the ¼ kg of groundnuts, chop the onions and tomatoes.
- (iv) Once the matoke is soft, mix in the groundnuts, onions, and tomatoes, and let it cook until the mixture is thick and well combined.

Preparing posho at home

Materials needed:

- ½ kg of maize flour
- water
- mingling stick
- saucepan

Steps taken:

- (i) Boil water in a saucepan and gradually add maize flour to the boiling water while stirring continuously to avoid lumps.
- (ii) Reduce the heat and keep stirring until the mixture thickens.
- (iii) Cover the saucepan and let it simmer on low heat for about 10-15 minutes, stirring occasionally.
- (iv) Remove the posho from heat and let it sit covered for a few minutes before serving.

Demonstration:



Preparing sweet potatoes

Materials needed:

- Sweet potatoes (at least 5 big tubers)
- water
- banana leaves
- knife
- saucepan
- banana stalks

Steps taken:

- (i) Wash and peel the tubers, then cut them into halves.
- (ii) Line the saucepan with banana stalks and banana leaves.
- (iii) Place the sweet potatoes on the banana leaves, add some water, and cover them with more banana leaves.
- (iv) Place the saucepan on the charcoal stove and cook until sweet potatoes are soft and ready to eat.

Demonstration:



Preparing boiled rice at home

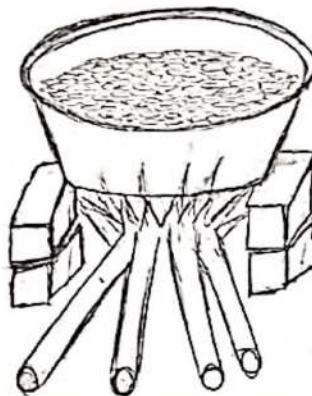
Materials needed:

- ½ kg of rice
- mingling stick
- 1 litre of water
- saucepan
- 1 litre of cooking oil
- charcoal stove

Steps taken:

- (i) Measure ½ kg of rice and rinse it with clean water.
- (ii) Pour 1 litre of water into the saucepan and place it on the charcoal stove to boil.
- (iii) Once the water is boiling, add the rinsed rice into the saucepan.
- (iv) Cover the saucepan and let the rice cook on low heat till all the water is absorbed and the rice is soft.

Demonstration:



Preparing millet/cassava bread at home

Materials needed:

- ¼ kg of millet flour
- water
- mingling stick
- saucepan
- ½ kg of cassava flour

Steps taken:

- (i) Pour half of a litre of water into a saucepan and place it on a charcoal stove to boil
- (ii) Add the ¼ kg of millet flour to the boiling water and stir it using the mingling stick.
- (iii) Slowly add the half kg of cassava flour, stirring continuously to avoid lumps.
- (iv) Keep stirring till the mixture thickens, then remove the saucepan from the stove, let it cool, and serve.

Demonstration:



Reasons for cooking food.

↓ To make it soft.

↓ To make it easy to digest

↓ To kill germs in food.

↓ To make it tasty.

Eating habits practised by people.**Good eating habits**

- Chewing food with mouth closed.
- Having meals at regular times.
- Putting small lumps of food in the mouth at a time.
- Sitting upright when eating.
- Eating food while quiet.
- Avoid eating in between meals.

Bad eating habits

- Eating food with unwashed hands.
- Talking with food in the mouth.
- Chewing food with the mouth open.
- Eating while walking.
- Bending while eating food.
- Swallowing big lumps of food at a time.

Dangers of bad eating habits.

❖ It causes choking.

❖ It causes stomach pain.

❖ It causes indigestion.

Evaluation activity 5.7

1. (a) Write down any **two** methods that can be used to prepare simple dishes at home.
 - (b) State any **two** steps that can be used when preparing local dishes at home.
 2. (a) Name any **two** local dishes that people commonly prepare at home using local food.
 - (b) Give any **two** reasons why people cook food at home.
 3. The sentences below show steps used when preparing posho at home in wrong order.
 - (i) Cover the saucepan and let it simmer on low heat for about 10-15 minutes stirring occasionally.
 - (ii) Remove the posho from heat and let it sit covered for a few minutes before serving.
 - (iii) Boil water in a saucepan and gradually add maize flour to the boiling water while stirring continuously to avoid lumps.
 - (iv) Reduce the heat and keep stirring until the mixture thickens.
- Write the sentences above in the spaces below by arranging them correctly to show the steps used when preparing posho.

(i)

.....

(ii)

.....

(iii)

.....

(iv)

.....

MEANING OF KEY TERMS IN OUR FOOD:

Food: A substance that adds nutritional value when taken into the body.

Carbohydrates: Energy giving food.

Balanced diet: A meal containing all food values in their right amount.

Food preservation: A way of keeping food safe for a long time without going bad.

Food hygiene: General cleanliness of food and things used to prepare it.

Malnutrition: A condition when the body does not receive essential nutrients.

Deficiency diseases: Diseases that spread due to poor feeding.

Proteins: Body building foods.

Vitamins: Health giving food.

Food values: Nutritional and dietary components of food.

Pickling: It involves soaking food in vinegar to preserve it.

Smoking: It is the exposure of food to slow burning heat to preserve it.

Diet: Refers to the overall pattern of food a person consumes over a period of time.

Meal: A single sitting of food eaten at specific time.

Food security: A situation in which there is enough food available all the time for consumption.

Foodborne diseases: Infections caused when a person eats contaminated food.

Food contamination: The presence of harmful substances in food that can make it unsafe to eat.

Nutrients: Substances which are essential for healthy growth.

Nutrition: The process by which food is taken in and used by the body.

Goitre: The swelling of the thyroid glands due to lack of iodine.

Anaemia: A condition in which the body lacks blood in it.

6

THEME: HUMAN BODY

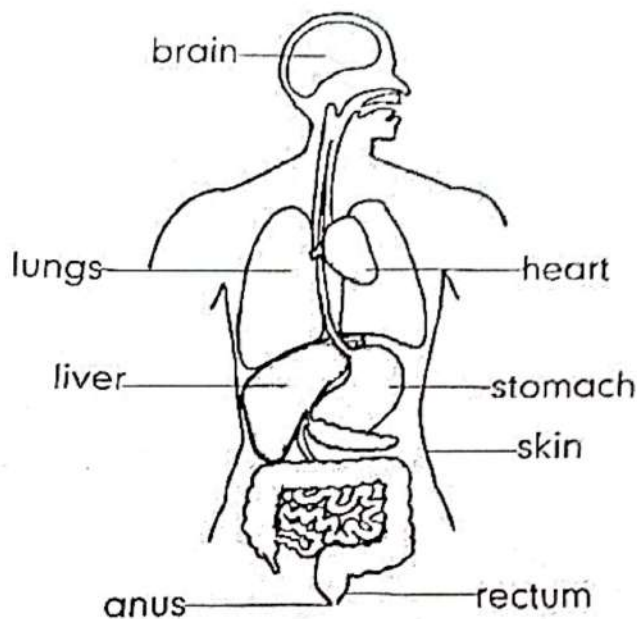
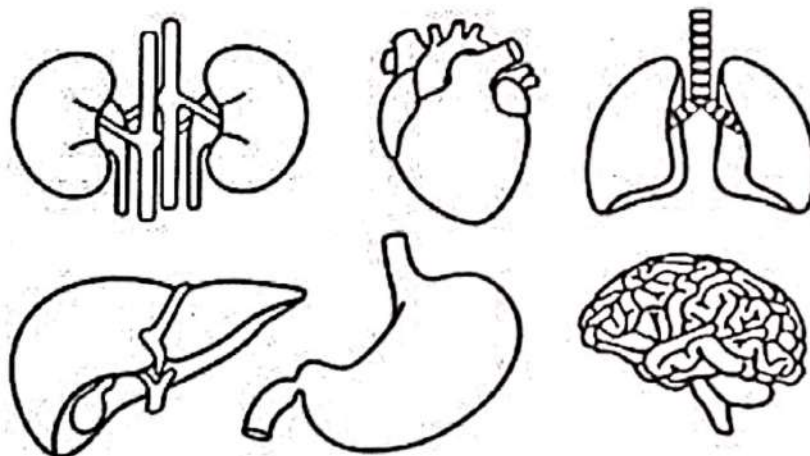
TOPIC 6: HUMAN BODY ORGANS

Key words:

- nasal • cilia • trachoma • orbit • conjunctivitis
- tongue

Major Organs of the Human Body

An organ is a group of different tissues which work together to do a particular function in the body.

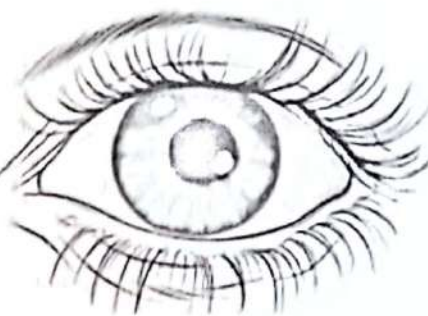
**Human body organs****The major human body organs include;**

- the eyes ➤ the brain ➤ the ears ➤ the stomach ➤ the heart
- the urinary bladder ➤ the lungs ➤ the liver ➤ the tongue
- the nose ➤ the kidneys.

The External Human Body Organs

The eyes

- The eye is the sense organ for seeing/sight.
- The eyes help us to identify the sizes, height and colours of objects in the environment.
- The eyes use light energy for them to function.
- The human eye is protected by the orbit (eye socket)



Diseases that attack the human eye

Night blindness, river blindness, trachoma, conjunctivitis and sty.

Night blindness is caused by lack of vitamin A in a diet. It is the deficiency disease that attacks the eyes.

River blindness is spread by a black fly.

Trachoma is spread by a housefly and caused by bacteria.

Conjunctivitis is caused by bacteria or virus.

Stye is an inflammation on the human eye due to bacterial invasion.

The disorders of the eyes

Short sight, long sight, blindness, old age sight and squint

Ways of caring for the eyes

- Regular washing of the eyes with clean water and soap. It helps to remove rheum from the eyes.
- Do not touch the eyes with dirty hands.
- Avoid looking at the sun directly.
- Avoid reading under too dim or too bright light.
- Having regular optical check-ups.

The nose.

The nose is the sense organ for smelling.

The nose has holes called nostrils.

The nose has cilia and mucus which filter air and trap dust.

Diseases of the nose

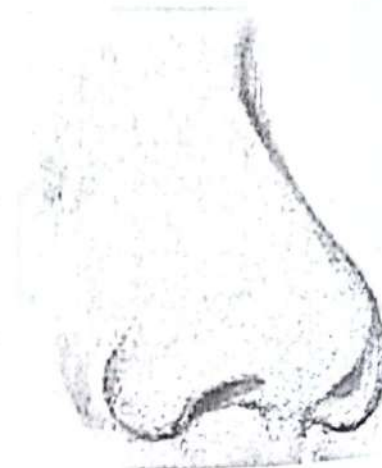
Sinusitis, nasal allergies and influenza

Disorders of the nose.

Nose bleeding, nasal congestion and foreign bodies in the nose

Care for the nose

- Avoid pushing objects into the nose.
- Removing mucus regularly.
- Wearing a face mask.
- Regular washing of the nose.
- Avoid staying in dusty places.
- Avoid rough games like slide kicks.
- Clean the nose using a clean handkerchief.



Face mask

The ear

The ear is the body organ for hearing.
The ear uses sound energy for it to function.
The ear is also used for body balance.
The human ear has wax which traps dust in the human ear.

Diseases of the ears.

Earache, otitis, ear cancer and otalgia

Disorders of the ears

Ear boils, ringing in the ear, deafness, itching inside the ear.

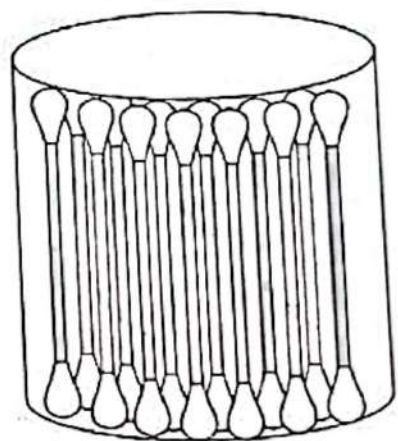
Care for the ears

- Going for regular medical check-ups.
- Clean the ear using ear cotton swabs.
- Avoid playing rough games like slide kicks.
- Listen to music at a low volume.
- Insert earplugs in the ear to protect the ear from loud noise.

Learn, unlearn and relearn: Ear buds are small earphones inserted in the ear for listening music thus are not used to clean the ear. Additionally, ear pads are cushioned parts of headphones or earphones or headsets that rest against or around your ears.



Ear cotton swabs



Evaluation activity 6.1

1. Name human body organ that requires light energy for it to function.
2. Apart from hearing, give **one** other function of the ear to a person.
3. Write down any **one** vector that affects the human eyes.
4. State the difference in function between the human nose and the eyes.
5. Give any **one** reason why it is dangerous to clean the ears using sharp objects.

The diagram below shows a material used to clean the human ears. Use it to answer questions 6 and 7.



6. Identify the material used to clean the human ears shown above.
7. Why is it advisable to use the above item once when cleaning the human ears?
8. Apart from the nose, mention any **one** other external organ of the human body.
9. State the importance of the orbit on the human body.
10. State the importance of the hair in the nose of a person.

Key words:

▪ brain ▪ taste

▪ bile

▪ regulate

▪ cirrhosis

▪ churn

The Internal Human Body Organs**The brain.**

The brain is protected by the skull.

The brain organises different organs to work together as a whole body.

The brain controls all other body organs.

**Functions of the brain**

- It stores information.
- It controls all body activities.
- It is used for body balance.
- It is used for remembering/recalling/memory.
- It is used for thinking.
- It is used for reasoning.
- It is used for learning new things.
- It is used to interpret information.
- It is used for decision making.

Diseases that attack the brain

Cerebral malaria, meningitis, epilepsy, brain cancer

*Note: Amnesia is a disorder of the brain that leads to memory loss.***Care for the brain**

- Have enough rest and sleep. It helps to refresh the brain.
- Feed on a balanced diet.
- Do regular physical exercises.
- Immunise children against meningitis.
- Avoid tobacco smoking and alcoholism.
- Early treatment of malaria.

The stomach

The stomach is part of the digestive system.

Chemical digestion of proteins begins from the stomach.

Digestion is the breakdown of food into simple soluble particles that the body can absorb.

**Functions of the stomach**

- It stores food for some time.
- It churns food.
- It absorbs alcohol.
- It produces hydrochloric acid that kills germs in food.
- It produces gastric juice.

Diseases that affect the stomach.

Peptic ulcers, gastritis and gastric cancer

Care for the stomach

- Limit alcohol intake.
- Do regular physical exercises

- Eat food in time.
- Limit intake of fatty food.

The tongue

It is found in the mouth.

Functions of the tongue

- ❖ It used for tasting.
- ❖ It is used in talking.
- ❖ It rolls food into bolus.
- ❖ It mixes food with saliva.

Diseases of the tongue

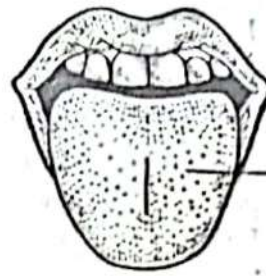
Oral herpes, oral thrush and oral cancer

Disorders of the tongue

Burns, cuts, blisters, taste loss

Care for the tongue

- Do not eat too hot food.
- Brush the tongue daily.
- It destroys taste buds.*
- It removes food remains from the tongue.*
- Do not put sharp objects in the mouth.
- Rinse the mouth to remove leftover food particles.
- Avoid smoking tobacco.
- Use a tongue scraper to remove dead cells and food remains.



The tongue

The liver

The liver is a reddish-brown organ in the body.

It is the largest internal organ in the human body.

Functions of the human liver.

- ✓ It regulates blood sugar.
- ✓ It produces bile juice.
- ✓ It stores food and blood.
- ✓ It removes poisonous substances from food and blood.

Diseases that affect the human liver.

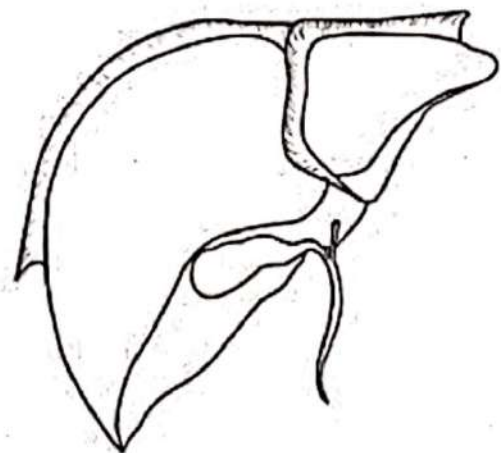
Hepatitis B, yellow fever, liver cancer, liver cirrhosis, liver abscess and hepatitis A. Hepatitis B, and yellow fever are immunisable diseases.

Hepatitis A is a waterborne immunisable disease

Liver cirrhosis is caused due to alcohol consumption.

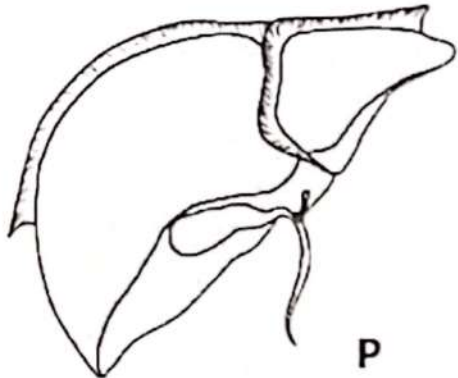
Care for the liver

- Feeding on a balanced diet.
- Doing regular physical exercises.
- Going for regular medical check-ups
- Avoid alcohol consumption.



Evaluation activity 6.2

1. Name the human body organ which helps a person to know whether there is enough sugar in tea or not.
 2. State **one** way in which the function of the brain is similar to that of the ear.
 3. Give any **two** ways of caring for the stomach.
 4. Which immunisable disease attacks the human liver?
 5. Name the body organ that requires one to have enough rest and sleep.
- The diagrams below show human body organs. Use them to answer the questions that follow.



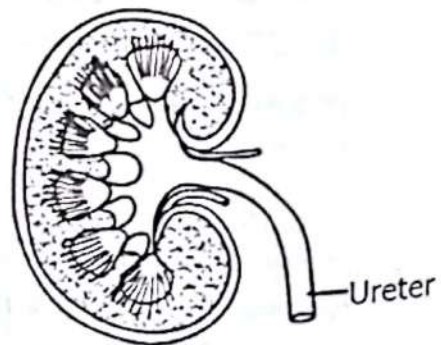
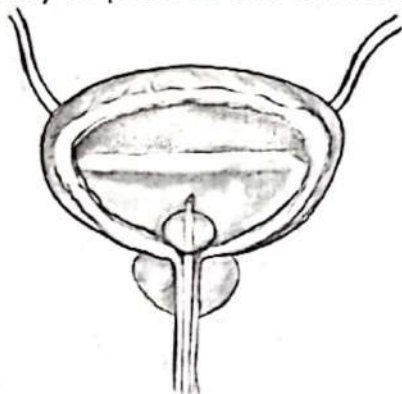
- (a) Name the body organs marked **P** and **Q**.
 - (b) Give any **one** function of the organ marked **P** in the human body.
 - (c) Which human body system does the organ marked **Q** form?
6. State **one** way in which it is important to have enough rest and sleep.
 7. How is the skull useful to the brain?
 8. Name **one** disease that affects the stomach because of alcoholism.

Key words:

- Bladder
- Vapour
- Urine
- Excretory
- Respiratory
- Influenza

The kidney

The kidney is bean-shaped and reddish brown in colour.
 The kidney works hand in hand with the urinary bladder.
 The urinary bladder stores urine for some time.
 The kidney is part of the excretory system.



The ureter takes urine to the urinary bladder.

Functions of the human kidney.

- It filters blood.
- It balances the amount of water in the body.
- It regulates salt levels in the body.

Diseases that affect the kidneys

Kidney stones, kidney cancer, kidney failure, bilharzia and gonorrhoea

Care for the kidneys.

- ⇓ Feed on a balanced diet.
- ⇓ Do regular physical exercises.
- ⇓ Drink enough boiled water.
- ⇓ Empty the urinary bladder when full.
- ⇓ Limit salt intake.
- ⇓ Going for regular medical check-ups.

The lungs

The lungs are used for gaseous exchange.

Lungs are used for breathing.

The lungs are protected by the rib cage.

Diseases of the lungs

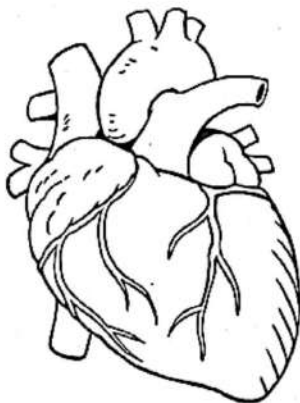
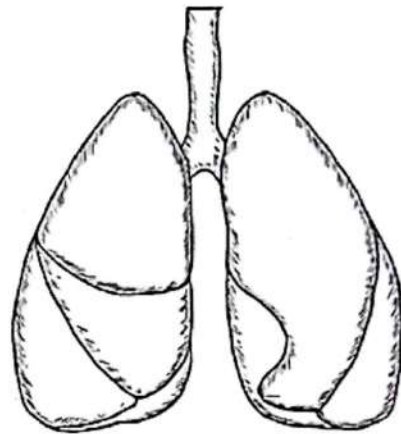
Lung cancer, tuberculosis, influenza, diphtheria, whooping cough, asthma and pneumonia.

Lung cancer is caused by tobacco smoking.

The heart.

The heart pumps blood in the body.

The heart is protected by the rib cage.



The human Heart

How the Human Body Works

How the body works	System used
Takes in food and oxygen. The processes involved are digestion, blood circulation and breathing.	• Digestive system • Respiratory system • Circulatory system
Carries food and oxygen to different parts of the body.	• Respiratory system • Circulatory system
Uses food and oxygen to get energy. This process is called respiration.	• Respiratory system • Circulatory system
Removes wastes. The process is called excretion.	• Excretory system

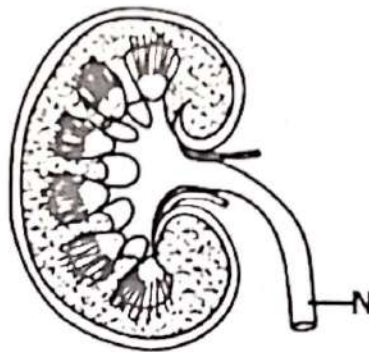
Examples of wastes removed from the body

Wastes	Organ that removes it	Process
Urine	Kidney	Urination
Sweat	Skin	Sweating
Carbon dioxide	Lungs	Respiration
Bile pigments	Liver	Excretion
Water vapour	Lungs	Respiration

Note: sweating cools the body. Respiration enables the body to get energy.

Evaluation activity 6.3

1. Name the human body organ that balances the amount of water needed in the human body.
 2. State the main function of the heart in the human body.
 3. Name the excretory organ that removes water and carbon dioxide from the human body.
 4. Name any **one** human body organ protected by the rib cage.
 5. By what process does the body produce energy from food and oxygen?
- The diagram below shows the human body organ. Use it to answer questions 6 and 7.



6. Name the organ shown above.
7. Give the function of the part labelled **N**.
8. The sentences below show how the human body works in wrong order.
 - (a) Carries food and oxygen to different parts of the body.
 - (b) Removes wastes.
 - (c) Takes in food and oxygen.
 - (d) Uses food and oxygen to get energy.

Write the sentences above correctly in the spaces provided below to show how the human body works.

.....

.....

.....

.....

Evaluation activity 6.3 Continued

9. The table below shows human body organs, the system to which they belong and the organ function. Study and complete it correctly.

ORGAN	SYSTEM	ORGAN FUNCTION
Heart	Circulatory	
.....	Excretory	Formation of urine.
Pancreas		Produces pancreatic juice.
Lungs	Respiratory	

10. Write down any **two** examples of wastes that are removed from the human body.

MEANING OF KEY WORDS IN HUMAN BODY ORGANS

Organ: A group of tissues performing the same function.

Brain: The human body organ for storing and interpreting information.

Skin: The human body organ for feeling.

Lungs: The human body organs for gaseous exchange.

Heart: The human body organ that pumps blood.

Kidney: The human body organ that filters blood.

Liver: The largest internal human body organ.

Stomach: The bag-like structure in which food from the stomach collects.

Body wastes: The metabolic materials removed from the body.

Digestion: The process by which food is broken down into smaller particles.

Excretion: The removal of metabolic waste materials from the human body.

Sweating: The process of removing sweat from the body.

Urination: The removal of urine from the human body.

Respiration: The breakdown of digested food to produce energy.

Breathing: The intake or outtake of air by an organism.

7

THEME: HUMAN BODY

TOPIC 7: THE TEETH

Key words:

• canine • biting • incisor • chisel • tear • permanent

Teeth are hard bone-like structures in the mouth of an animal.
Teeth help to break down food into small particles for easy swallowing.

Sets of teeth

Milk teeth and permanent teeth.

Milk teeth

These are the first set of teeth to grow in the mouth.

In a child, they are 20 teeth.

At around 5 years of age, these teeth begin to fall out.

This is called teeth shedding.

Milk teeth are also called deciduous teeth or primary teeth.

Dental formula for milk teeth

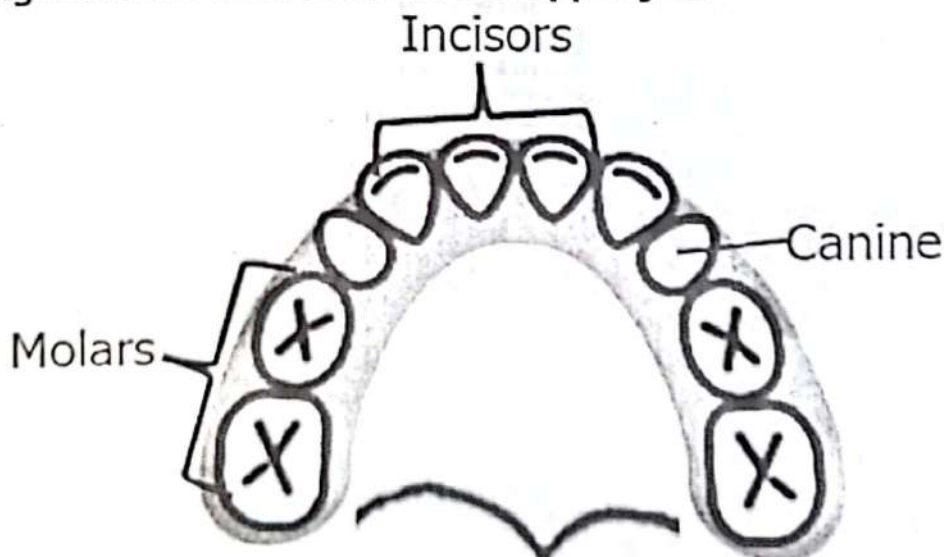
Type of teeth	Upper jaw	Lower jaw	Total teeth
Incisors	4	4	8
Canines	2	2	4
Premolars	0	0	0
Molars	4	4	8

Total in one jaw = 10 teeth

Total in the mouth = 20 teeth

Missing teeth: premolars

Arrangement of milk teeth in the upper jaw



Points to note:

Premolars are missing in milk set because they are not needed for grinding food at an early age, as a baby primarily consumes soft or liquid diet.

On the other hand, molars are present in milk teeth to help grind and chew soft solid foods as the baby transitions to a varied diet, unlike premolar, which are more specialised and develop later to support the stronger chewing needs of permanent set.

Permanent teeth

These are the second and the final/last set of teeth to grow in the mouth of a person.

They are also called secondary teeth.

An adult person has 32 permanent teeth in the mouth.

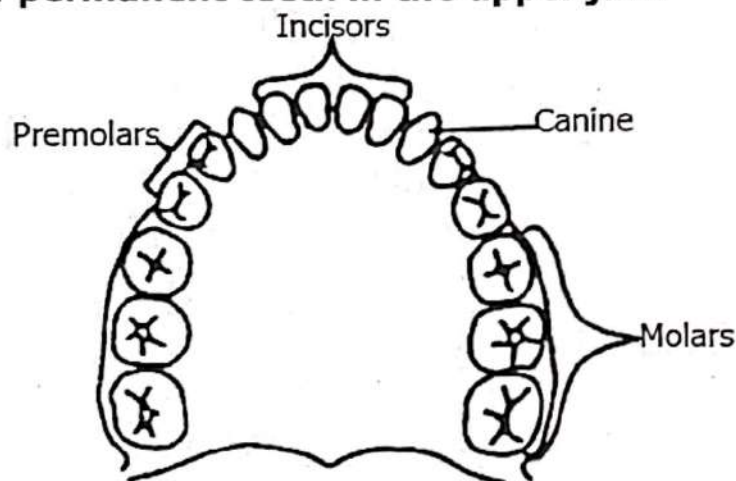
Permanent teeth begin developing at the age of 13 years.

Dental Formular in Permanent Teeth

Type of teeth	Upper jaw	Lower jaw	Total teeth
Incisors	4	4	8
Canines	2	2	4
Premolars	4	4	8
Molars	6	6	12

$8+4+8+12 = 32$ teeth.

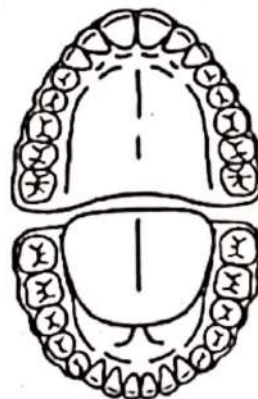
Arrangement of permanent teeth in the upper jaw.



Arrangement of milk teeth and permanent teeth in the mouth



Milk teeth



Permanent teeth

A SIMPLE METHOD TO ENABLE YOU REMEMBER THE TYPES OF TEETH IN EACH SET

A teacher asks learners these questions and then they give you answers.

1. When Ian is a baby, can he manage to pick mangoes from a tree?

Then you reply, No, Ian can't manage (ICM), So, no picking, no premolars.

2. When Ian is old, can he pick mangoes from a tree?

Then you reply, Yes, Ian can pick mangoes. (ICPM)

In summary, the method is;

Milk set: Ian can't manage (ICM)

Permanent set: Ian can pick mangoes. (ICPM)

In brief, the milk set lacks premolars.

Types of Teeth and their Functions

Type of teeth	Characteristics	Function and adaptation
Incisor teeth	- First type of teeth to grow in human beings and first to fall out. - They are chisel shaped. - They have straight edges. - They have one root.	For cutting or biting food. Adaptation. They are chisel-shaped to cut food.
Canine teeth	- They are sharp and pointed on top. - They have one root. - They are cone-shaped. <i>Note: They are found at the corner of the mouth.</i>	For tearing flesh. Adaptation. They are sharp and pointed to tear flesh
Premolar teeth	- They have blunt broad ridges. - They are smaller than molars. - They have one root. - They have cusps.	For crushing, grinding and chewing food. Adaptation - They have blunt broad ridges to grind and chew food. - They are broad to increase the surface area for chewing food. - They have cusps for chewing food.

Type of teeth	Characteristics	Function and adaptation
Molar teeth	- They are the last type of teeth to grow in human beings. -They have blunt broad ridges. -They are bigger than premolars. -They have two or three roots. Molars in lower jaw have two roots while those in upper jaw have one root -They have rough blunt tops. -They have many cusps and ridges. -They are the largest teeth.	For crushing, grinding and chewing food. Adaptation They have blunt broad ridges to grind and chew food. -They are broad to increase the surface area for chewing food. -They have cusps for chewing food.

Incisor



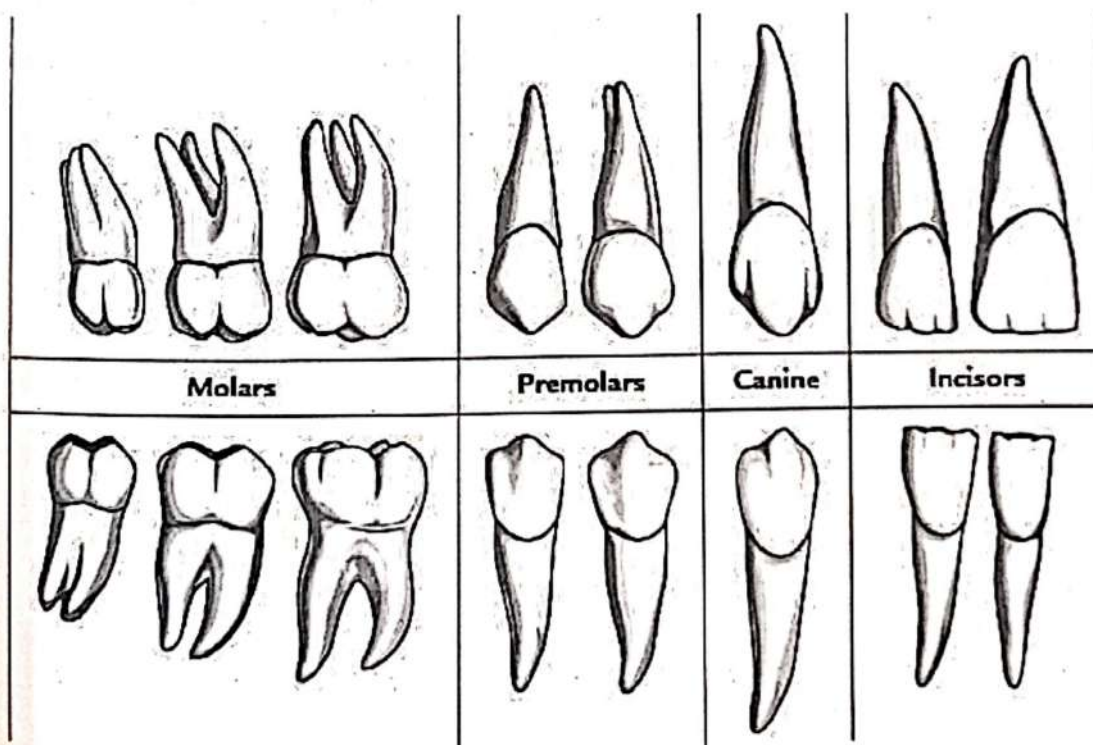
Canine



Premolar



Molar





A girl biting a yellow banana



A boy tearing meat.

Points to note:

Incisors are well developed in rats, squirrels, moles, mice and rabbits.

Canines are well developed in dogs, cats, lions, leopards and foxes.

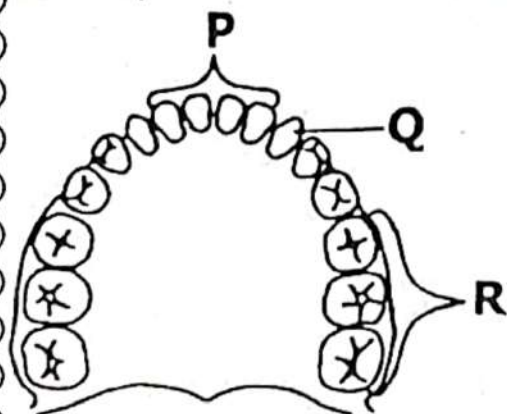
Evaluation activity 7.1

1. Name the type of teeth which develops last in human beings.
2. State any **one** characteristic of incisor teeth which makes them suitable for their function.
3. Name the type of teeth that has;
 - (i) Sharp pointed crowns.....
 - (ii) Chisel shaped crowns.....
4. The diagrams below show types of teeth. Use them to answer the questions that follow.



- (a) Name the types of teeth labelled **J** and **K**.
- (b) Give the function of the type of tooth labelled **J**.
- (c) State **one** way in which the type of tooth marked **K** is able to carry out its function during feeding.

5. The diagram below shows the arrangement of permanent teeth in the lower jaw of a human being. Study the diagram and use it to answer the questions that follow.

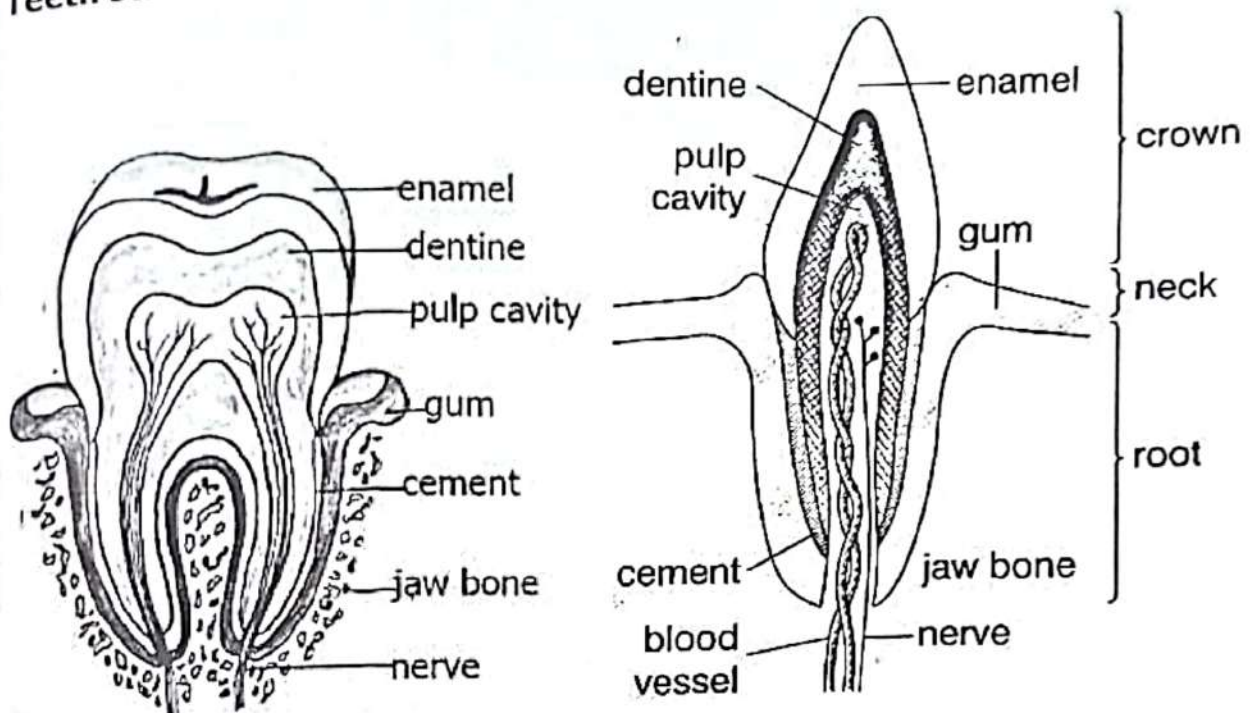


- (a) Name the type of teeth marked **Q**.
- (b) State the function of the types of teeth labelled **P** and **Q**.
- (c) Identify any **one** difference between the structure of the types of teeth labelled **Q** and **R**.

Key words:

- enamel • capillaries • phosphorus • dentine • calcium
- pulp

Teeth structure



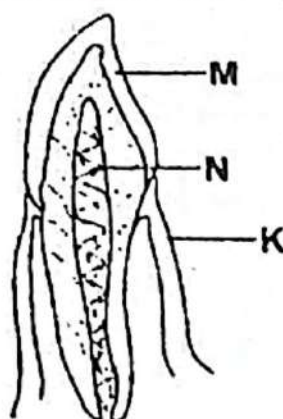
Functions of each part of the tooth.

Part	Description and function
Enamel	• Enamel is the hardest part of the tooth/substance in the body. • It covers the crown of the tooth. • It is the outer most part of the crown. • It is made up of calcium and phosphorus. • It is the white part of the tooth. • It protects the inner parts of the tooth. • It prevents wear and tear of the tooth. Adaptations to its function. - It is hard which makes it resistant to wear and tear. - It has smooth surfaces which prevent bacteria from easily sticking onto it. - It is thick to withstand chewing and biting forces.
Dentine	▪ It is the largest part of the tooth and contains living cells. ▪ It is covered by both the enamel and cement. ▪ It protects the pulp cavity. ▪ It reinforces the enamel and supports the structure of the tooth. ▪ It helps in the formation of the enamel.

Part	Description and function
Pulp cavity	• It contains blood capillaries and nerve endings. • Blood capillaries supply food and oxygen to the tooth • It is the most sensitive part of the tooth. • It contains living cells. It is the living part of the tooth.
Cement	▪ It holds the tooth firmly into the jawbone. ▪ It covers the root of the tooth.
Jawbone	• It holds the tooth in position. • It gives extra support to the tooth in the jawbone.
Gum	▪ It contains fibres for holding the tooth in the jawbone. ▪ It covers the jawbone.
Nerves	• Nerves are sensitive to pain, heat or cold. • They carry electrical messages about heat and pain to the brain.

Evaluation activity 7.2

1. Name the part of the human tooth that contains blood capillaries and nerve endings.
2. State **one** function of the enamel in the human tooth.
3. Name the part of the tooth that covers the dentine in the following regions.
 - (i) Crown
 - (ii) Root
4. Identify the hardest part of the human tooth.
5. State the difference in function between the pulp cavity and the cementum in the tooth.
6. Give the reason why the pulp cavity is regarded as the living part of the tooth.
7. The diagram below shows a section through a human tooth. Study and use it to answer the questions that follow



- (a) Name the parts marked **M** and **N**.
- (b) Which mineral salt helps to form part **M**?
- (c) Give the function of part labelled **N**.

Evaluation activity 7.2 Continued

8. The table below shows parts of the human tooth in part **A** and their functions in part **B**.

Part A: Parts	Part B: Function
Jawbone	Sends impulses of pain to the brain.
Cementum	Holds the tooth in position.
Dentine	Replaces the enamel when it wears off.
Nerves	Fixes the tooth firmly in the jawbone.

Write the correct function to the part of the tooth in the spaces provided below.

- (i) Jawbone
 (ii) Cementum
 (iii) Dentine
 (iv) Nerves

Signs/indicators of dirty teeth

- Bad breath.
- Bleeding gums.
- Tartar build-up.
- Dental plaque.
- Yellow stains.
- Discolouration.

Care for Our Teeth**❖ Brushing teeth regularly.**

It helps to remove food remains that attract bacteria in teeth.

Brushing teeth is recommended in the morning and every after a meal.

It is done using a toothbrush, toothpaste, charcoal paste and chewed stick.

Importance toothpaste

- Tooth paste prevents bad breath
- Toothpaste prevents tooth decay.
- Toothpaste removes plaque or stains from teeth.
- Toothpaste makes teeth strong.
- Toothpaste kills germs in teeth.



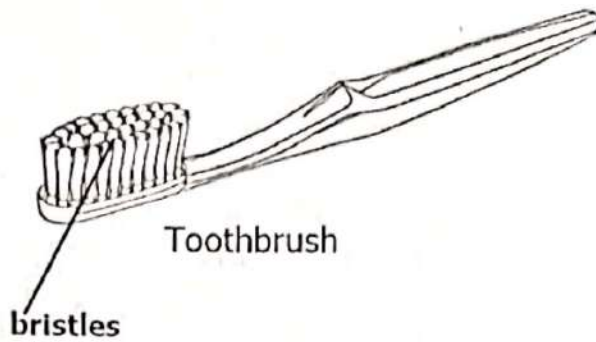
Toothpaste



Charcoal paste



Chewed stick



Steps taken when brushing teeth

- i. Wet the toothbrush and apply a small amount of toothpaste.
- ii. Brush your teeth using gentle circular motions, covering all surfaces-front, back and chewing surfaces.
- iii. Brush your tongue to remove food remains.
- iv. Rinse your mouth and toothbrush thoroughly with water.

Reasons for brushing teeth.

- ✓ To remove food remains that attract bacteria in teeth.
- ✓ To prevent bad oral breath.

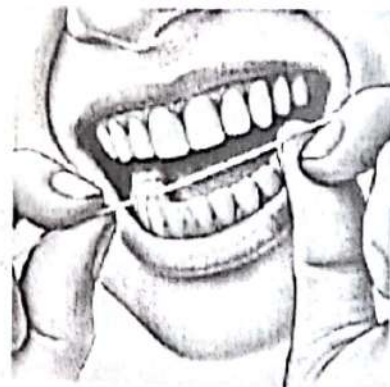
❖ Dental flossing

Dental flossing is the removal of plaque that forms between teeth using a thread like material.

It helps to prevent gum disease.

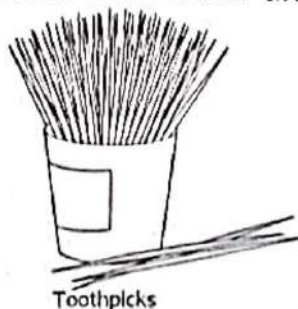


Dental floss



Dental flossing

- ❖ Using toothpicks to remove food remains from teeth.
- ❖ Rinsing the mouth with water and salt. Salt kills bacteria.

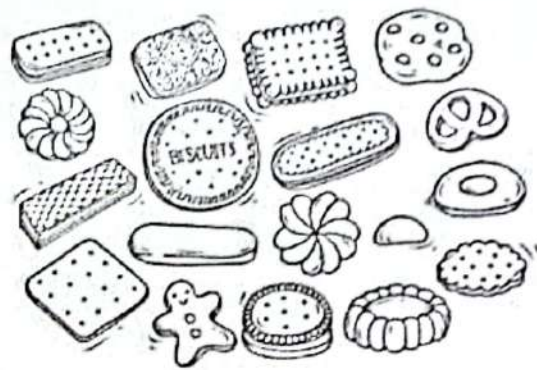


Toothpicks

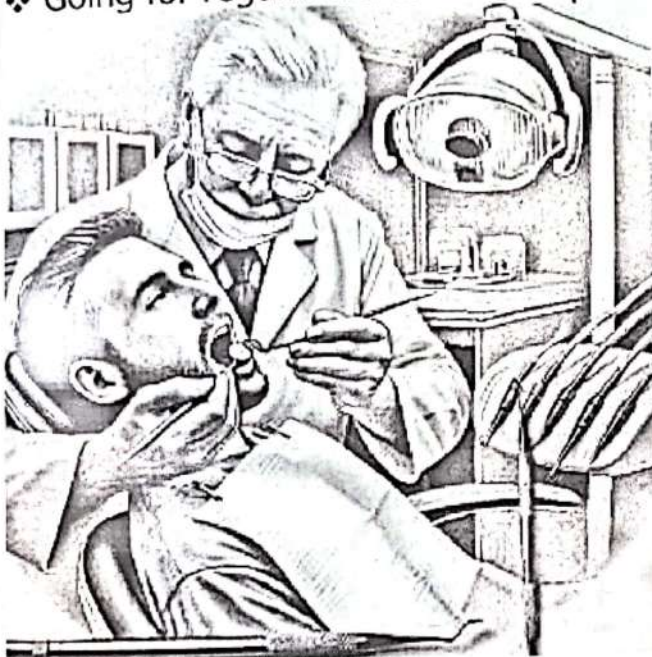


Rinsing the mouth.

- ❖ Eating correct food.
- ❖ Avoid eating very hot and cold food. This weakens the enamel.
- ❖ Avoid eating too much sweets and biscuits. They attract bacteria which cause tooth decay.



- ❖ Going for regular dental check-ups.



Advantages of regular dental check-ups.

- It helps to detect oral health issues early.
- It helps to maintain fresh breath and clean teeth.
- It helps to reduce the risks of gum disease.
- ❖ Avoid opening hard objects with teeth.
It causes breaking or cracking of the enamel.
- ❖ Feeding on plenty of fruits and vegetables to clean the mouth.

- ❖ Avoid tobacco smoking.

Tobacco smoking causes; Bad oral breath, Tooth discolouration, Build-up of plaque and Gum disease and oral cancer.

Evaluation activity 7.3

1. Name any **one** material that can be used to brush teeth.
2. State any **two** ways in which toothpaste is important when brushing teeth.
3. Give **one** reason why children are discouraged from eating too much sweets.
4. Write down any **two** practices that can destroy teeth in one's mouth.
5. Mention any **two** ways in which tobacco smoking affects teeth in the mouth.
6. State any **one** reason why children are encouraged to brush their teeth regularly.

Evaluation activity 7.3 Continued

The diagram below shows materials used to clean the teeth. Use it to answer questions 7 and 8.



6. Name the materials shown above.
7. Give any **one** way in which the above materials help to clean teeth.

Teeth Diseases and Disorders.**Tooth decay (dental caries)**

It is caused by bacteria.

It is characterised by dental cavity in the teeth.

Foods like chocolates, cakes and sweets attract bacteria that cause tooth decay.



A tooth affected with dental caries

Gum disease (gingivitis)

It is caused by bacteria. It results from poor dental hygiene.

Signs of gum disease

- Swollen gums.
- Bleeding gums when brushing teeth.
- Bright red gums.
- Bad breath from the mouth.

Root abscess

This is a pus-filled infection at the root of a tooth.

It is caused by bacteria.

Signs and symptoms of root abscess

- Severe toothache.
- Swelling.
- Fever
- Sensitivity to hot or cold food.

Periodontitis

This is a severe gum infection that damages the soft tissue and bone-supporting teeth.

It is caused by bacteria.

Signs and symptoms of periodontitis

- Swollen gums.
- Bleeding while brushing.
- Bad breath.
- Loose teeth.

Use mnemonic: GACP to recall teeth diseases

Ways in which diseases that attack teeth can be controlled/prevented.

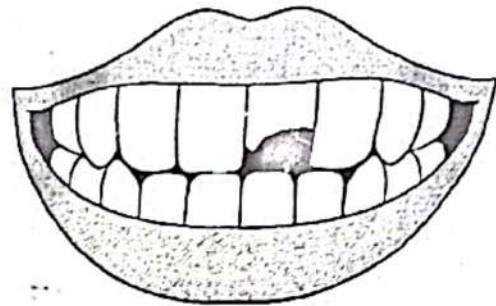
- Regular and proper brushing of teeth.
- Limit the intake of sugary food.
- Regular dental check-ups.
- Dental flossing.
- Avoid smoking.
- Use antibacterial mouthwash.

Dental Disorders

Broken or cracked teeth

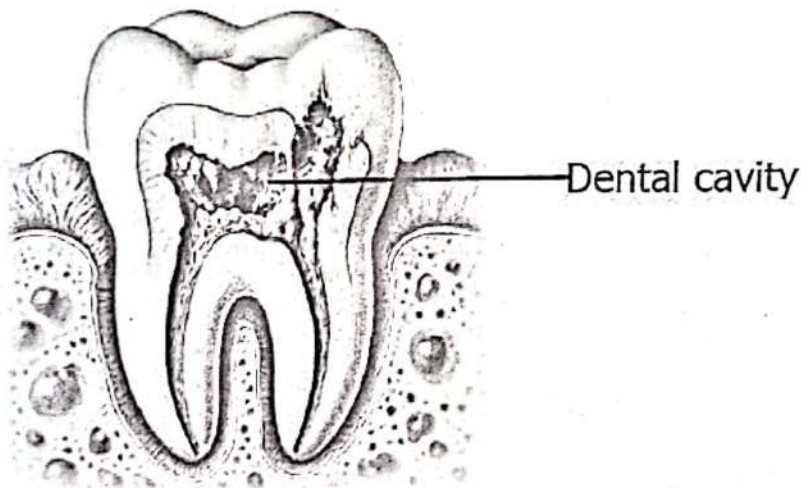
Causes of broken teeth are;

- Falls
- Little calcium
- Opening bottles using teeth
- Eating hard food stuffs like sugarcane.



Dental cavities.

These are holes in the tooth which are caused by bacterial infection. The holes can be dental filled using dental amalgam.



Dental plaque

This is a sticky film of food debris, saliva and bacteria which form on teeth after meals.

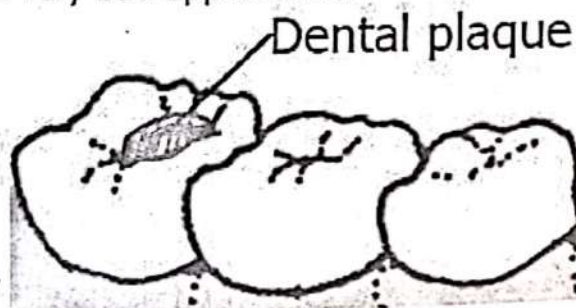
Dental plaque forms **tartar** when not removed by brushing.

Tartar is a hard yellowish or brownish substance on teeth caused by plaque.

Dangers of tartar on teeth.

It leads to gum disease and tooth decay.

It gives teeth a very bad appearance.

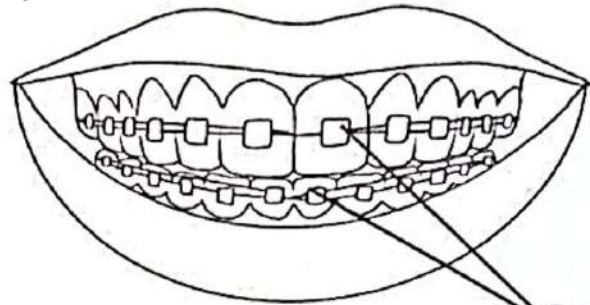
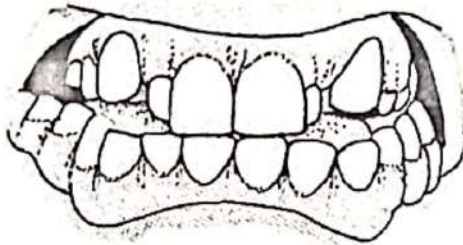


Improper growth of teeth

This is a dental condition in which teeth grow in a wrong way.

Causes of improper growth of teeth.

- Dental accidents.
- Mouth breathing.
- Fingernail biting.
- Lip biting.
- Thumb sucking.
- Early loss of milk.



Dental brace

A dental brace is used to correct misaligned teeth

Signs of improper growth of teeth

- Too much space between teeth.
- Missing teeth.
- Misaligned teeth.
- Some teeth grow on top of others.

Dangers of improper growth of teeth

- Difficulty in chewing.
- Dental diseases can easily develop.
- Poor facial appearance.
- Speech problems.

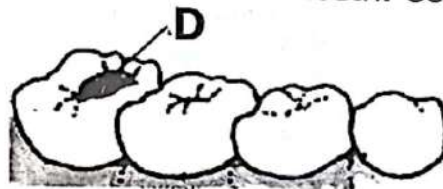
Other dental disorders

- False teeth.
- Discolouration of teeth.
- Malocclusion (Misalignment of teeth in the jaws)
- Impacted teeth.
- Bruxism (Grinding teeth)

Evaluation activity 7.4

1. Name the germ that causes dental caries.
2. State any **one** characteristic of incisor teeth that makes them suitable for their function.
3. State any **one** way in which toothpaste helps to promote oral health.
4. State any **two** signs of gum disease in humans.
5. Write down any **two** diseases that attack human teeth.

The diagram below shows ill health on teeth. Use it to answer questions 6 and 7



6. Name the substance marked **D**.
7. How are the above teeth able to grind food?
8. State the importance of sharp pointed crowns to the canine teeth.
9. Apart from brushing, give any **one** other way of preventing dental caries.

MEANING OF KEY WORDS IN TEETH

Enamel. The hardest part of the tooth.

Dental. Relating to teeth.

Tartar. Hard deposit formed on teeth and not removed by brushing.

Dentist. A doctor who treats teeth.

Dentistry. The care and treatment of teeth, gums and oral health.

Canines. Sharp cone-shaped teeth used to tear food.

Incisors. Sharp chisel-shaped teeth at the front of the mouth used to cut food.

Molars. Blunt broad ridged teeth with larger surface area used to grind and chew food.

Dental formula. The number and types of teeth in one's mouth.

Dentition. The arrangement of teeth in the jaws.

Oral health. The general cleanliness of the mouth.

Malocclusion. The misalignment of teeth in the jaws.

Dental plaque. A substance containing bacteria that forms on teeth surface.

Dental cavity. Holes in the teeth.

8

THEME: HUMAN HEALTH TOPIC 8: SANITATION

Sanitation is the general cleanliness of the environment.
It is the way of keeping an area clean.

Activities that school children can do to promote sanitation

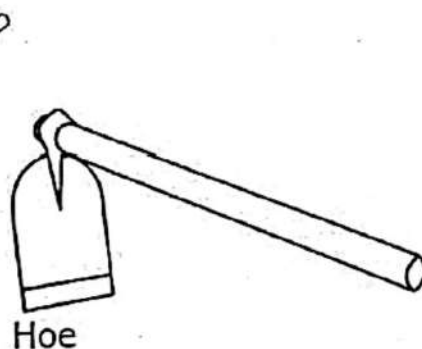
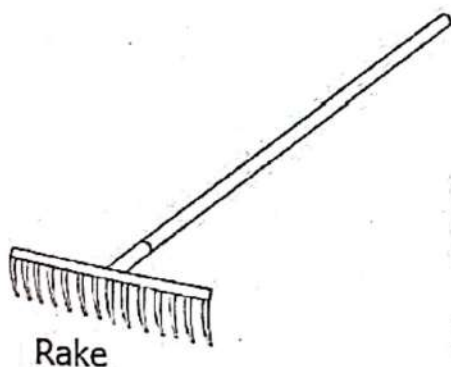
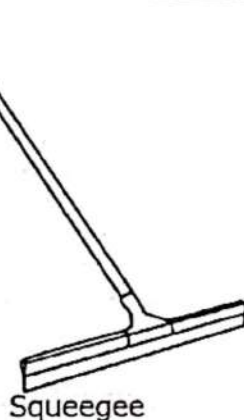
- ✓ Mopping classrooms.
- ✓ Sweeping the compound.
- ✓ Draining stagnant water.
- ✓ Smoking latrines.
- ✓ Scrubbing toilets.
- ✓ Proper disposal of human wastes.
- ✓ Dusting tables and chairs.
- ✓ Clearing bushes around the school.
- ✓ Scrubbing verandas and floors.
- ✓ Constructing urinals.
- ✓ Flushing toilets after use.
- ✓ Proper disposal of rubbish



Materials used to promote sanitation.

Material	Use in promoting sanitation
water	For mopping houses For scrubbing toilets/latrines/paved compounds.
Broom	For sweeping rubbish in the house or compound
Rag/mop	For mopping the floor to remove dirt.

Material	Use in promoting sanitation
Slasher	For cutting tall grass in the compound.
Liquid soap	For mopping and scrubbing the floor and toilets. For disinfecting toilets/latrines
Squeegee	For drying water on floor after mopping.
Floor brush	For scrubbing dirt on the floor.
Detergents	For mopping, scrubbing and disinfecting the floor.
Rake	For collecting rubbish in the compound.
Hoe	For trimming the compound.
Dustpan	For collecting rubbish in the house after sweeping.



Learn, unlearn and relearn: A squeezer is not used for drying the floor. It is used for making passion fruit juice. Additionally, the steel wire is not used for scrubbing saucepans and other metallic utensils. Steel wool is used for making window frames, door frames. The fibre like material used for scrubbing saucepans is called steel wool.

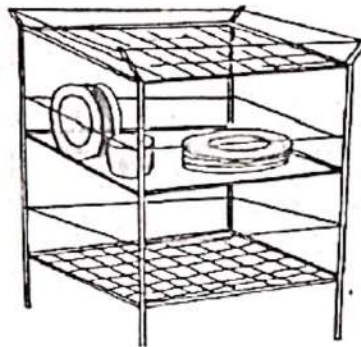
Element	Importance/use
Clean latrines/ toilets	For proper disposal of human wastes. Note well. A latrine is a structure built on a pit for disposing human wastes. A toilet is a waterborne facility for disposing human wastes.

Element	Importance/use
Clean bathrooms	For bathing
Rubbish pit/dust-bin	For proper disposal of rubbish.
Plate stand/rack	For proper drying of utensils and cutlery.
Tippy-tap/hand washing facility	For washing hands.
Animal houses.	For sheltering farm animals.

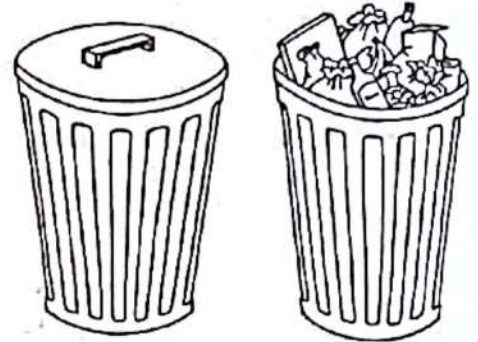
Hand washing facility



Plate stand



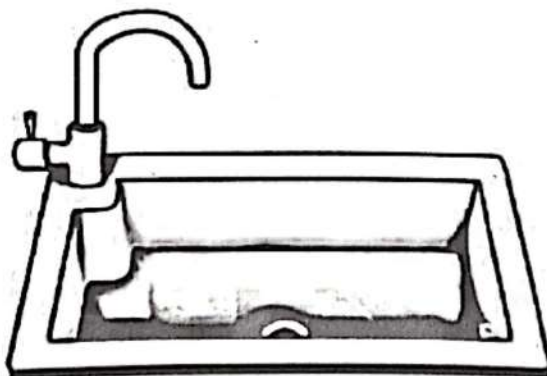
Dustbins



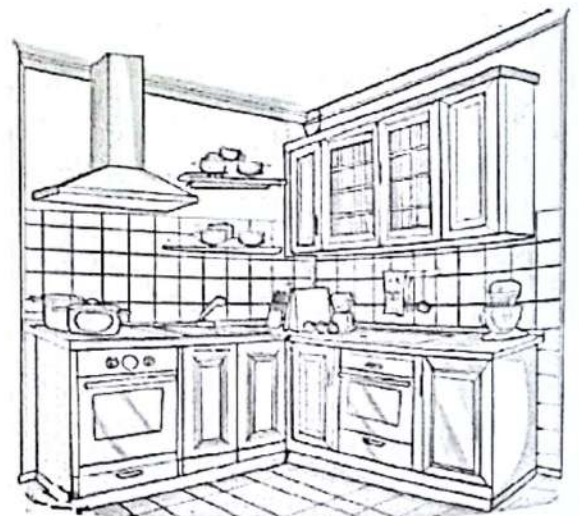
Flush toilet



Pit latrine



Wash basin/sink



Kitchen

Don't Speak for quality let quality speak for itself

Points to note.

 The stand has to be raised at least 1 metre above the ground

Reasons

- To prevent animals and young children from dirtying washed utensils and cutlery.
- To prevent water splashes from dirtying utensils in case it is raining.
- To prevent dust from dirtying washed utensils.

A sink is used for washing utensils and cutlery.

A dustbin should have a lid

Reasons.

To prevent rubbish from coming out.

To prevent bad smell from the dustbin from coming out.

To prevent easy spread of germs by houseflies.

Places at home and school that need to be kept clean regularly

- Kitchen ➤ Latrines/toilets ➤ Houses ➤ Compound
- Classrooms ➤ Bathrooms ➤ dormitories

Indicators of poor sanitation

- ✓ Bushy compound. ✓ Dusty classrooms.
- ✓ littered compound. ✓ Pools of water in the compound.
- ✓ Poor disposal of faeces, rubbish, domestic wastes.
- ✓ High number of houseflies in the compound.

Dangers/disadvantages of sharing the house with domestic animals

- It leads to the spread of ectoparasites.
- It leads to competition for oxygen.
- It leads to bad smell in the house.
- It leads to spread of some diseases.
- Animals can cause noise pollution in the house
- Animals destroy household

Dangers of poor sanitation in an area.

- ⇕ It leads to easy spread of diarrhoeal diseases.
- ⇕ It leads to water and air pollution.
- ⇕ It leads to contamination of food by houseflies.
- ⇕ It encourages the breeding of vectors and germs.

Importance of promoting proper sanitation

- ❖ It prevents easy spread of diarrhoeal diseases.
- ❖ It prevents air and water pollution.
- ❖ It reduces the breeding of disease vectors and germs.
- ❖ It controls bad smell in the environment.

Messages that can be displayed in the school compound to promote sanitation at the school.

- Do not litter the compound.
- Keep the compound clean daily.
- Flush the toilets after use.
- Dump rubbish in the dustbins/rubbish pits.
- Empty dustbins when full.
- Report leaking hand washing facilities.
- Use the toilets/latrines properly.

Activities a School Health Club can do to promote sanitation at school.

- Providing dustbins at school.
- Providing handwashing facilities.
- Organizing smoking of pit latrines.
- Conducting talks on sanitation practices
- Digging a rubbish pit at school.
- Creating posters and signs having sanitation messages.

Evaluation activity 8.1

1. (a) Give the meaning of the term sanitation.
- (b) Identify any **two** activities that school children can do to promote sanitation at school.
- (c) Mention any **one** place at school that needs to be kept clean.
2. The diagram below shows a sanitary item. Study and use it to answer the questions that follow.



- (a) Name the sanitary item shown above.
- (b) How is the above item useful in the promotion of sanitation at home?
- (c) State any **two** ways of caring for the sanitary item above.
- 3.(a) State any **two** indicators of poor sanitation in a school.
- (b) Give any **two** diseases that can break out due to poor sanitation in a school.
- 4.(a) Give **two** reasons for placing the utensils on a rack.
- (b) Why should a rack be constructed with stands at least 1 metre above the ground?
- (c) Give any **one** danger of leaving utensils outside for a long time on the rack.
5. The diagram below shows a hand washing facility. Study and use it to answer the questions that follow.



- (a) Name the hand washing facility shown above.
- (b) How does the above hand washing facility help to prevent the spread of diarrhoeal diseases?
- (c) What is the importance of the stick on the above hand washing facility?
- (d) State any **one** way of keeping the hand washing facility above clean.

Key words

• inhale • microscope • protozoa • hairy • in luenza • microbe

GERMS AND DISEASES

Germ is an organism that causes diseases.

Germ is also called **pathogens**.

Most germ is microbe/microscopic because they cannot be seen using our naked eyes.

A microscope is used to see germ.

It is commonly used by health workers to see germ in blood, stool and saliva.

This enables them to give us the correct treatment.

Characteristics of germ.

- ❖ They cause diseases.
- ❖ Most of them are microbes.
- ❖ They reproduce faster.

Examples/types of germ

Germ	Diseases they cause
bacteria	tuberculosis typhoid scabies cholera
fungi	ringworm infection athlete's foot dhobi's itch jock itch
viruses	AIDS influenza polio measles
protozoa	malaria sleeping sickness
Worms	bilharzia, river blindness and elephantiasis



A microscope

Places where germ can be found.

- soil
- air
- latrines/toilets
- rubbish pits
- rotting matter
- unboiled water

Ways in which germ spread.

- Through drinking unboiled water.

Clean water is not safe water. Clean water still contains germ.

- Through body contact.

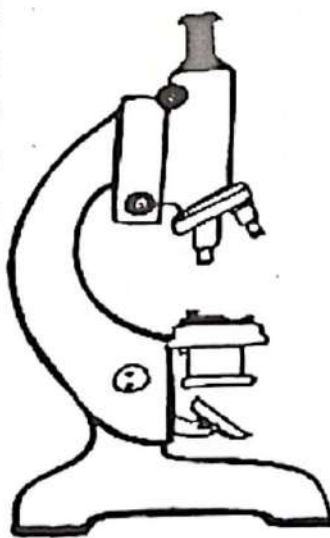
Body parts through which germ can spread are skin, hands, mouth and shoulders.

- Through bites of insects or dogs.
Such insects include; mosquitoes, tsetse flies, ticks and lice.
- Through inhaling contaminated air.
Bacteria and viruses can spread through air.
Diseases like influenza, COVID 19, tuberculosis, whooping cough and mumps can spread through inhaling contaminated air.
- Through sharing skin cutting objects.
Some germs are found in blood. Objects like razorblades, needles, syringes and safety pins can spread germs in the body.
Medical cutting or piercing equipment are sterilised to kill germs. Alcohol is mostly used.
- Through sharing clothes, combs, towels and bedding.
Fungi and bacteria can easily spread through sharing such materials.
Ringworm and scabies can spread fast when such materials are shared.
- Through the hairy body of some insects.
Houseflies and cockroaches have a hairy body on which germs attach.
Hairs are sticky for germs to easily attach onto them.

Evaluation activity 8.2

1. Name the group of organisms that cause diseases in human beings.
2. (a) What is meant by the term germs?
(b) Apart from bacteria and viruses, name any **two** other examples of germs.
(c) State any **two** ways in which human beings contract germs.
3. (a) List down any **two** places where germs can be found.
(b) State any **two** characteristics of germs.
4. How is a housefly able to spread germs?
5. Write down any **two** diseases that can spread through air.

The diagram below shows a medical equipment. Study and use it to answer questions **6** and **7**.



6. Name the medical equipment shown above.
7. How is the medical equipment shown above useful to a doctor?
8. State any **one** danger of sharing stockings with your fellow pupils at school.
9. Why are germs called microscopic organisms?
10. Mention any **one** body fluid where germs can be found.
11. Why should health workers sterilise medical cutting equipment before using them to other patients?
12. Name any **two** insects that spread germs through bites.
13. How do cockroaches spread germs to people?

Key words

▪ fingers ▪ faeces ▪ hairy ▪ diarrhoeal ▪ disposal ▪ rubbish

THE 4Fs GERM PATH

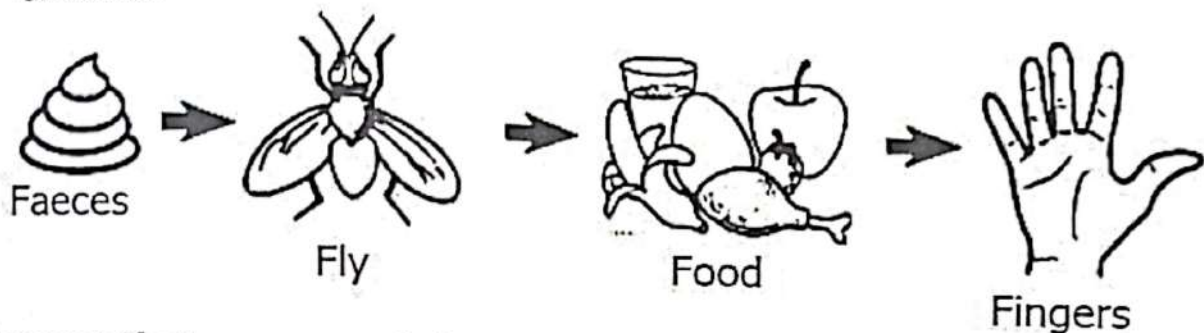
4Fs germ path refers to the routes through which houseflies spread germs that cause diarrhoeal diseases to humans.

A housefly has a hairy body to carry germs.

The 4Fs in germ path stands for Faeces, Flies, Food and Fingers.

How to explain the 4Fs germ path.

- (i) Germs breed in faeces.
- (ii) Flies pick germs from faeces using their hairy bodies onto the food.
- (iii) The food is then contaminated by the germs put by the houseflies.
- (iv) The person uses fingers to eat the contaminated food causing diarrhoeal diseases.



Diseases that are spread through the 4Fs germ path.

These diseases are called diarrhoeal diseases.

Examples of diarrhoeal diseases are; diarrhoea, dysentery, typhoid and cholera.

Ways in which the spread of germs can be controlled in an area.

- ❖ Covering leftover food properly.
It prevents houseflies and cockroaches from landing on food.
It prevents dust from getting on food.
- ❖ Collecting and burning rubbish.
It destroys the breeding of houseflies and cockroaches.
Houseflies lay eggs in decaying matter.

Reasons.

- To enable their larvae get food after being hatched.
- Presence of warmth for eggs to hatch.
- Presence of moisture for the eggs to hatch.
- ❖ Washing hands with clean water and soap.
To remove and kill germs from hands.
- ❖ Serving, cooking and keeping food in clean places and containers.
- ❖ Drinking boiled water.
Boiled water is free from germs.
Boiling water kills germs.

Ways in which rubbish can be disposed in an area.

- ✓ Burning rubbish.
- ✓ Burying rubbish.
- ✓ Wearing face masks.
- ✓ Dumping rubbish in dustbins or rubbish pits.

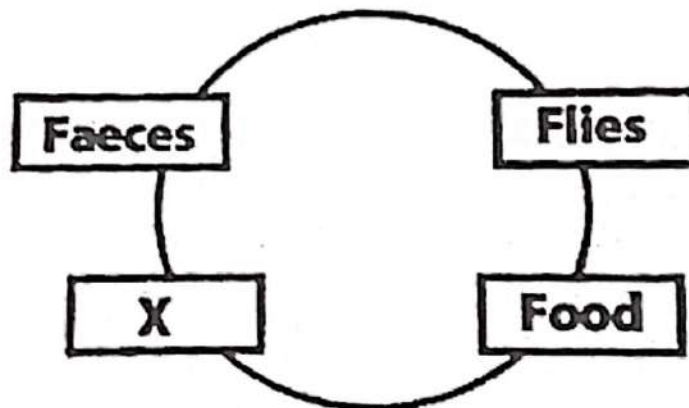
- ✓ Using decayable rubbish in production of biogas.
- ✓ Using decayable rubbish in making manure.

Learn, unlearn and relearn: It is wrong to say that we wash our hands with clean water to kill germs because water has no chemicals which kill germs on hands. It is only soap which kill germs on hands. Therefore, we wash hands with clean water, to remove germs from hands.

Additionally, we wash our hands with clean water and soap to remove and kill germs from the hands.

Evaluation activity 8.3

1. Write down any **two** of the 4Fs in the transmission of diseases.
2. What type of diseases are spread through the 4Fs germ path?
3. Which vector is active in the 4Fs germ path?
4. State any **two** ways in which germs can spread through the 4Fs germ path.
5. How is a housefly able to spread germs to human beings?
6. The diagram below shows how germs can spread through the 4Fs. Study and use it to answer the questions that follow.



- (a) Identify the word which should be in the box **X**.
- (b) State any **two** activities that can be done to prevent the germ path shown in the diagram above.
- (c) What structure on the body of flies in the germ path above enables it to carry germs onto food?
7. What health habit should you do to after visiting a latrine/toilet?
8. Why do houseflies lay eggs in decaying matter?
9. Write down any **two** diseases that can spread through the 4Fs germ path.
10. How does proper covering of food control germs spread by houseflies?

Germs cause rotting

Rotting is the breakdown of dead organic matter by bacteria or fungi. It is also called decomposition or decaying.

The fungi that cause rotting are called **moulds**.

It requires warmth and moisture.

Organic materials that can rot are; food, animal dung, paper, faeces, plant leaves, dead bodies

Importance of rotting.

- ✓ It produces humus from dead organic matter.
- ✓ It destroys garbage heaps.
- ✓ It destroys faeces in latrines and sewage systems.

Note. It is not advisable to pour acid in latrines and septic tanks.

Reason. Acid kills bacteria that cause rotting of faeces.

Dangers of rotting.

- ⇅ It causes wounds to become septic.
- ⇅ It is a source of germs.
- ⇅ It makes food to go bad.
- ⇅ It attracts vectors.

Signs of rotting on food.

- Growth of moulds on food.
- Change in colour.
- Bad smell.
- Presence of maggots

Ways in which rotting can be prevented in an area.

- Heating food.
- Smoking food.
- Eating food immediately after cooking.
- Putting food in refrigerators.
- Proper sun drying of food.

Ways of preventing diseases without using drugs.

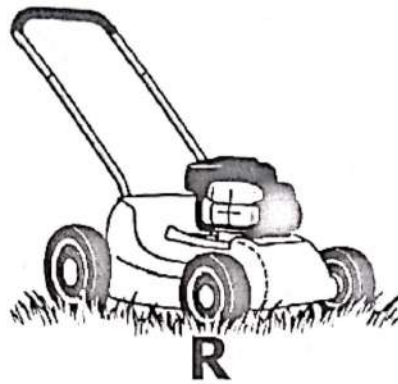
- Feeding on a balanced diet.
- Having enough rest and sleep.
- Doing regular physical exercises.
- Avoid smoking and alcoholism

Evaluation activity 8.4

1. What is meant by the term **rotting**?
2. State any **one** condition favourable for rotting to take place.
3. Name any **one** organism that causes rotting.
4. Give any **two** ways in which rotting is useful in the environment.
5. State any **two** dangers of rotting in the environment.
6. Why is it not advisable to pour acid in latrines or septic tanks?
7. Write down any **two** substances that can rot in the environment.
8. State any **two** changes on food that show that it is rotting.
9. Give any **two** ways of preventing diseases without using essential drugs.
10. Identify the process that reduces the volume of faeces in latrines and septic tanks.
11. State any **two** ways of preventing rotting of food at home.
12. Name the fungus that normally grows on leftover food.
13. In **four** sentences, describe an experiment to show rotting of a ripe banana fruit.

Evaluation activity 8.4 Continued

13. The diagrams below show equipment used to promote sanitation at home. Study and use them to answer the questions that follow.



- Name the equipment marked **R** and **S**.
- State the importance of the equipment marked **S** in the promotion of sanitation at home.
- Give any **one** reason why most homes do not use the equipment marked **R** to promote sanitation.

MEANING OF KEY WORDS IN SANITATION

Sanitation. The general cleanliness of the environment.

Germs. Organisms that cause diseases.

Mower. A device used to cut grass in fields or lawns.

Rag. A piece of old cloth used for mopping.

Microscope. An instrument used to see germs.

Latrine. A place for urination and defecation.

Mop. An implement that consists of a bundle of thick loose strings or a sponge attached to a handle, used for wiping floors.

Excreta. Human wastes especially urine and faeces.

Rotting. The breakdown of organic matter by bacteria or fungi.

Squeegee. A tool for washing floors, that has a long handle with two thick pieces of soft materials at the end used to remove water or other liquids from the floor.

Environment. The surrounding of an organism.